

Tax Withholding and the Size of Government^{*}

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Abstract

This paper examines the hypothesis that an improvement in tax collections causally leads to bigger government. We exploit the staggered introduction of withholding of the state personal income tax by U.S. states between 1948 and 1987 and find that withholding led to an increase in tax revenues by about 28 percent. We derive a theoretical model through which we interpret the estimates distinguishing between a mechanical increase in tax collections driven by improved compliance, subsequent adjustments in revenue choices in response to that improved compliance, and an increase in the underlying demand for revenue that may have motivated the adoption of withholding. Governments responded to the improvement in personal income tax collections by shifting the composition of revenues towards a heavier reliance on this tax. States also increased tax rates as they implemented withholding, which suggests that a need to raise more revenue was an important motive for adopting withholding.

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JEL classification codes: H11, H21, H26, H71, N42.

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“Withholding may make it easier to pretend you aren’t being taxed but it’s also easier for government to raise taxes without getting a protest from the people.” – Ronald Reagan, 1967¹

1 Introduction

Explaining the growth of government during the 20th century has been a vexing question in economics. While the growth in the size of the national government in a number of developed countries is well-known and has been analyzed (e.g., Tanzi and Schuknecht, 2000), perhaps less well-known is the growth in the size of sub-national governments during the 20th century. But as evidenced by Figure 1, the size of U.S. state governments, as measured by the ratio of state expenses to personal income, more than trebled from around 4 percent at the end of World War II to over 12 percent by the year 2000. Moreover, during this period state personal income tax revenues also progressively became a more important source of revenue, increasing more than four-fold from 0.5 percent of personal income in 1944 to approximately 2.75 percent by the year 2000. In this paper, we examine this significant growth in the size of state government and focus on one possible explanation that has been offered for the growth of government in other contexts, viz. the adoption of a more efficient mode of tax collection: employer withholding of the personal income tax.

The literature examining causes for the growth in the size of government is voluminous and we do not attempt to summarize it all here. We simply note that one strand of the literature argues that governments became bigger mainly because they improved their capacity to raise taxes, and to collect them more efficiently (e.g., Kau and Rubin, 1981, and Becker and Mulligan, 2003). In fact, practically all positive models of the size of government reach the same prediction: an improvement in the government’s ability to collect taxes should increase the size of government.² Other authors however argue that a growing demand for public spending has been the primary driving force behind the growth of government. In this view, the causality runs in the opposite direction: bigger governments consciously build the capacity to collect large amounts of revenue as the benefits of doing so are greater (e.g., Wallis, 2000, and Lee, Kim, and Borchering, 2013).

A very different body of literature in public economics investigates administrative mechanisms

¹<https://www.sethkaller.com/item/1567-24387-Governor-Ronald-Reagan-Opposes-Withholding-of-State-Income-Tax&from=12>

²The relevant models include the Becker and Mulligan (2003) model of interest groups, the Meltzer and Richard (1981) median voter model of pure redistribution, the Hettich and Winer (1988) model of determination of tax structure under probabilistic voting, or the Brennan and Buchanan (1980) model of Leviathan government.

that enable modern governments to raise large amounts of revenue to begin with (e.g., Kleven, 2014, and Kleven, Kreiner, and Saez, 2016). In this line of research, third-party information reporting (e.g., Gordon and Li, 2005; Kopczuk and Slemrod, 2006; Kleven et al., 2011; and Adhikari et al., 2022) and tax withholding (e.g., Slemrod and Velayudhan, 2018; Brockmeyer and Hernandez, 2019; Bagchi, 2022; Bagchi and Dušek, 2021; and Waseem, 2022) play a key role in the ability of governments to collect tax revenue. These papers convincingly demonstrate a causal impact from the introduction of third-party information reporting and/or withholding on taxes collected through improved tax compliance. None of the papers in this literature, however, trace out the impacts of an improvement in tax collections on overall public finances, i.e. on the composition of taxes or on the overall level of government revenues and expenditures.

Our paper connects these two threads in the economics literature that have largely remained disconnected from each other. Exploiting the staggered introduction of withholding for the state personal income tax by U.S. states between 1948 and 1987, we investigate if an improvement in tax collections results in a larger government or whether causality runs in the opposite direction. Withholding, as we use the term in this paper, combines third-party reporting by which the revenue agency receives reports of the compensation paid to employees and the act of withholding itself, i.e. collecting the tax at the source rather than from the income recipient. As we establish in a previous paper (Bagchi and Dušek, 2021), withholding immediately and permanently increased personal income tax collections by 29 percent, holding tax rates constant.³ Such a major improvement in tax collections could (and, as our theoretical model predicts, should) have led to broader changes in the level and composition of tax revenues. Those adjustments, driven by the political responses to the improvement in tax collections, are the primary focus of this paper. We therefore add to the existing literature by studying the broader fiscal impacts of employer withholding on all major sources of state tax revenues, tax rates, as well as on expenditures and debt levels: effects that go beyond the mechanical increases in personal income tax revenues due to improved compliance.

Withholding of the personal income tax was adopted at the federal level in 1943 with the goal to raise more revenue to finance the war effort. The hypothesis that it caused further tax changes later on, and enabled the post-war expansion of federal personal income taxes and the federal government as such, was postulated by none other than Milton Friedman, who helped implement

³We further provided evidence that this increase in collections was driven by higher compliance rather than changes in the tax base or higher levels of enforcement.

withholding when he worked for the U.S. Treasury at the time. As he stated in his memoirs:

[W]ithout a system of current collection, it would have been impossible to collect the amount of income taxes that we collected during the war. At the time, we concentrated single-mindedly on promoting the war effort. We gave next to no consideration to any longer-run consequences. It never occurred to me at the time that I was helping to develop machinery that would make possible a government that I would come to criticize severely as too large, too intrusive, too destructive of freedom.⁴

Ascribing causality from withholding to the post-war growth of the federal government is fraught with empirical challenges. The staggered introduction of withholding at the state level, however, provides us a credible set-up for tracing the effect of an improvement in tax collection on the level and composition of tax revenues. The variation in the timing of adoption was substantial, starting with Oregon in 1948 and concluding with North Dakota in 1987 allowing us to use a difference-in-differences estimator with a state-level panel dataset. The interpretation of our estimates is further aided by the fact that withholding improved the ability of governments to collect the personal income tax, but not other taxes such as the general sales tax or excise taxes. Furthermore, as exemplified by the case study for California in the main text (and three additional case studies in Online Appendix B), the specific timing of adoption was a function of idiosyncratic political circumstances, often hinging on the personalities involved rather than the specific fiscal needs of the state.

To guide our empirical work, we build a simple model of the government's choice of the size of government and composition of tax revenues. We model withholding as an improvement in the ability of governments to collect the personal income tax but allow for the possibility that its adoption may be motivated by a higher demand for spending. The model distinguishes between a mechanical increase in income tax revenues due to improved compliance and subsequent political adjustments to all taxes in the new equilibrium. The model predicts that if withholding were adopted exogenously, then total revenues would increase, income tax revenues would increase, income tax rates would be reduced, and other tax revenues (and rates) decrease. Because it is

⁴Friedman and Friedman, p. 123.

optimal for governments to rely more heavily on the tax the ease of collecting which has improved, personal income tax revenues would increase relative to other taxes. On the other hand, if the adoption of withholding was motivated by a greater demand for spending, then revenues from other taxes (and tax rates) may, in fact, *increase*.⁵

We find that following the adoption of withholding, state governments started raising substantially more revenues from the personal income tax, by about 28 percent. There is some evidence that states also started raising more revenues through the corporate income and sales taxes but those increases are much smaller than the increase for the personal income tax. Finally, we also find that the top personal income tax rate and the corporate income tax rate *increase* by about 15 percent following the adoption of withholding. These findings are consistent with the model's prediction regarding the responses of the tax structure in the new equilibrium *and* the presence of a positive demand shock for tax revenues. Although state tax revenues across all sources increase significantly by about 5 percent following the introduction of withholding, there are no statistically significant increases in state expenditures; this apparent anomaly is explained by the fact that state debt levels stabilize once they adopt withholding. Overall, our evidence suggests that the demand for higher government revenues drove the adoption of withholding in the first place and that factor was quantitatively more important in explaining the increase in the size of government than any increase brought about because of adopting a tax system with higher compliance.

This paper contributes to the literature in several ways. First, as pointed out above, we estimate broader fiscal effects of withholding and third-party reporting, and thus extend the literature that has so far focused solely on the compliance mechanisms. Second, it provides evidence on the claimed causal link from an improvement in the ability to collect taxes to the size of government in a setting that allows for a more credible interpretation of the results than the previous literature. In fact, evidence on the hypothesis that an improvement in the government's ability to collect taxes leads causally to bigger government is rather scant and based primarily on cross-country work. Kau and Rubin (1981) argue that the costs of collecting taxes have declined because changes in the structure of the economy made tax collection easier. They find that factors such as the growth in female labor force participation and decline in self employment explain virtually all

⁵Total tax revenues and income tax revenues would continue to increase because of the added ease of collecting revenues (as before) but also because of the greater demand for spending that we characterize as a demand shock.

the growth of government during 1929–70.⁶ Becker and Mulligan (2003) find a positive relationship between various measures of tax efficiency⁷ and the size of government in a cross section of countries. Jensen (2022) documents a strong link between formalization⁸ and the tax capacity of governments, in a cross-section of countries and using a long-run time series for the United States (1870–2010) and for Mexico (1960–2010).

In contrast to these papers, Wallis (2000) finds little evidence to suggest that growth in the size of government since the late 19th century has been driven by a reduction in the cost of raising revenues and appears skeptical of the view that tinkering with revenue structure would bring about a fundamental change in the size of government. Likewise, Lee, Kim, and Borchering (2013) find, in a panel of 29 OECD countries over 1970–2007, that introductions of the value added tax (a relatively more efficient tax) had little effect on government growth and instead the demand for government spending was the primary determinant of the tax structure. The last two papers are relevant for the third contribution of this paper: its assessment of the relative contribution of the mechanical and demand factors. We find that although there were adjustments in the level and composition of taxes following the introduction of withholding, a demand shock likely motivated its adoption and played a more important role in the growth of *state* governments during the period 1944–2000. Unlike much of the literature that relies on cross-country variation, our conclusions are based on exploiting variation across U.S. states where differences in political institutions, cultural norms, and economic structure are much smaller compared to differences across countries.

To comment further on the relative contribution of the mechanical and demand factors, we attempt to quantify the degree to which the adoption of withholding was responsible for the growth of state income taxes. Our estimates suggest that withholding accounts for only about 11 percent of the overall growth of income tax revenues between 1944–2000. Given that the adoption of withholding was likely the single biggest improvement in tax administration in this period, we conclude that a growth in the demand for spending was a more important factor behind the rise of state income taxes rather than improvements in tax compliance. Consistent with these conclu-

⁶Quite plausibly though, the causality may also run from higher female labor force participation to higher demand for government spending.

⁷For example, the share of government revenue raised from relatively more efficient taxes, or the ratio of average to marginal income tax rate.

⁸Formalization is specifically captured by a higher employee share in the labor force and the share of employees above the personal income tax exemption threshold (thus, higher share of income potentially subject to third-party reporting and withholding).

sions, we find that although withholding resulted in a large *one-time* increase in personal income tax revenues, it did *not* change the growth rate of tax revenues by itself.

The rest of the paper is organized as follows. Section 2 provides institutional background on tax withholding at the state level. Section 3 develops a theoretical model and derives testable predictions. Section 4 explains our estimation strategy and Section 5 describes the data. Empirical results on taxes, tax rates, expenses, and debt are presented in Section 6. We address the contribution of withholding to the growth in the size of government in Section 7. Section 8 concludes.

2 Income tax withholding - institutional background⁹

Today, personal income taxes are the single most important source of revenue for state governments, representing 38 percent of states' total tax revenue.¹⁰ However, state income taxes were relatively unimportant until the 1950s. In 1911, two years before the federal income tax in its modern form was enacted through the Sixteenth Amendment, Wisconsin became the first state to enact a modern, broad-based income tax. State individual income taxes became more common during the 1930s when 18 states implemented them. State income taxes however generally tended to be narrow, levied mainly on very high income earners (Gillitzer, 2017), and even by 1944, they represented less than 9 percent of tax revenues in the 29 states that were imposing them, and only 5 percent across all states.

By introducing withholding of the personal income tax in 1943 the federal government made a major change in tax administration.¹¹ Until then taxpayers remitted their income taxes when they filed annual returns. Following the introduction of withholding however, employers were mandated to withhold a fraction of their workers' paychecks and remit it directly to the government. On their end, taxpayers were required to file income tax returns at the close of the tax year in which they computed the assessed tax. Any difference between the income tax assessed and the amount withheld during the previous year either had to be remitted to the government (if positive)

⁹The historical information in this section builds on Penniman (1980). Similar text appears in our complementary paper (Bagchi and Dušek, 2021), but we provide the key background information here for the reader's convenience.

¹⁰Source: Census of Governments (2022). General sales taxes and excise taxes contribute about 29% and 13% of total tax revenue respectively.

¹¹Withholding was used on a limited scale in connection with the income tax briefly imposed during the Civil War. Withholding reappears in the federal context in 1913 on the heels of the ratification of the Sixteenth Amendment but withholding was repealed in 1917 (Mehrotra, 2016).

or would be refunded to the taxpayer (if negative).

Some limited forms of withholding had existed at the state level even before 1943.¹² In 1948, Oregon became the first state to require employer withholding on all wage and salary income. Nine states adopted withholding in the following 10 years, five states in 1959, and 12 states adopted it during the 1960s (see Table 1). Barring North Dakota, all 29 states that had a broad-based income tax at the end of World War II introduced withholding by the early 1970s, with California introducing withholding in 1971. There were also 10 states that introduced a broad-based individual income tax during the 1960s and 1970s and all of these states chose to withhold from the outset suggesting that withholding had become an “industry standard” in tax collection. Notably, no state has abolished it.

Withholding, as the term is understood in the context of the state personal income tax, combined two innovations: third-party reporting (the revenue agency receiving reports of the compensation paid to employees) and withholding as such (collecting the tax from the source of the income rather than its recipient). It therefore has the potential to increase tax revenue primarily by improving tax compliance (Kleven, Kreiner, and Saez, 2016). Beyond that withholding raises tax revenue through more subtle channels. When taxes are progressive and incomes are rising, more taxpayers move into higher tax brackets and the average tax rate rises. As withholding shifts the tax collection forward, the government permanently receives the *growth* in average tax rate about half a year earlier than it would have under a traditional regime in which taxes are remitted at the end of the year. Lastly, it effectively increases government revenue by overwithholding (Hauck and Wallossek, 2024). Given that both the state and federal systems are designed such that the average taxpayer receives a refund, the government obtains an interest-free loan from taxpayers (Jones, 2012).

3 Theoretical model

We develop a simple model with a twofold objective: to inform the regressions that estimate the revenue effects of withholding and to disentangle the mechanisms through which revenue effects operate. The key idea is that withholding increases the revenue collected at given tax rates from one tax source (the personal income tax). Equivalently, it allows governments to raise a given

¹²For example, a few states required withholding of the non-residents’ income.

amount of revenue at lower marginal tax rates and therefore at lower marginal deadweight costs. This triggers a political adjustment in both the level and composition of tax revenues resulting in a new equilibrium.

We follow Samuelson (1954) and adopt the framework of a benevolent social planner in which political equilibrium results from maximizing a weighted sum of agents' utilities. The government provides public spending G , the social benefit of which, $\gamma B(G)$, is an increasing and concave function. The parameter γ reflects both the value of public spending to its recipients as well as their weight in the planner's problem. The government revenue comes from two sources, the personal income tax, R^I , and other taxes, R^O .

To collect the personal income tax, the government imposes a flat rate τ^I on income Y^I . Tax collection is imperfect, hence, the revenue actually collected is a fraction $a \in (0, 1)$ of the revenue under perfect compliance:

$$R^I = a\tau^I Y^I \quad (1)$$

Equation (1) defines the technology of personal income tax collection. The parameter a captures the state's tax capacity, influenced by, among other factors, the underlying economy (e.g, the fraction of formal employment), and the state's expenditures on tax enforcement. The term, aY^I , is the fraction of income that is effectively being taxed.

The planner trades off the benefits from spending against the deadweight costs of taxation. The deadweight costs of the personal income tax increase linearly in the effectively taxed income and the elasticity of taxable income, ϵ , and quadratically in the tax rate (following Feldstein, 1999):

$$C(R^I) = \frac{1}{2}(\tau^I)^2(aY^I)\epsilon = \frac{1}{2}\frac{\epsilon}{aY^I}(R^I)^2 \quad (2)$$

where the last expression follows from the substitution of equation (1). Equation (2) explicitly shows that the deadweight costs of the personal income tax fall with an improvement in tax collection.^{13, 14}

¹³If the state's revenue needs are exogenously determined such that the state needs to raise a fixed amount, R^I , regardless of the level of compliance, then equation (1) makes it clear that a higher level of a corresponds to a lower tax rate, τ^I , and deadweight costs decrease quadratically with a decrease in the tax rate. Thus, holding R^I constant, as compliance with the income tax improves, deadweight costs decline and vice versa.

¹⁴We briefly discuss an alternative formulation in which we consider $a\tau^I$ to be the effective tax rate and analyze the effect of a change in a on deadweight costs. In this alternative formulation, $C(R^I) = \frac{1}{2}(a\tau^I)^2 Y^I \epsilon = \frac{1}{2}\frac{\epsilon}{Y^I}(R^I)^2$ implying that the deadweight costs of the income tax are unaffected by the parameter a representing the fraction of the revenue being collected. While this formulation may have some intuitive appeal, the predictions resulting from such a formulation are inconsistent with our empirical observations. If the deadweight costs of the income tax were unresponsive to changes in a , then we would not find any change in the reliance of governments on the personal income

Similarly, the revenue from other taxes is a product of the tax rate τ^O and tax base Y^O :

$$R = \tau^O Y^O \quad (3)$$

The collection technology of other taxes is not explicitly modeled as it is held independent of the collection of the personal income tax and thus unaffected by withholding; similarly, the deadweight costs of other taxes are independent of the deadweight costs of the personal income tax¹⁵ and increase quadratically in revenue:

$$C(R^O) = \frac{1}{2} \frac{\delta}{Y^O} (R^O)^2 \quad (4)$$

While we use the technical term “deadweight costs”, the costs need not be limited to the traditional efficiency costs of taxes as often measured by the Harberger (1964) triangle. Instead, our formulation is much more general. Thus the parameters δ and ϵ can also capture other costs, such as the administrative costs of taxation and/or the political costs to elected officials from taxing various groups. The key element of the model is that the cost that the planner considers increases quadratically (at least beyond some point) and that the two parameters, δ and ϵ , determine the cost of each tax source.¹⁶

The government’s problem is to choose R^I and R^O in order to maximize the social benefit of spending net of the deadweight costs of taxes:¹⁷

$$\max_{R^I, R^O} \gamma B(G) - \frac{1}{2} \frac{\epsilon}{aY^I} (R^I)^2 - \frac{1}{2} \frac{\delta}{Y^O} (R^O)^2 \text{ s.t. } R^I + R^O = G \quad (5)$$

The solution to the problem is best characterized in two stages: In the first stage, for any given G , the government chooses the composition of taxes $(R^{I*}(G), R^{O*}(G))$ that minimizes the deadweight costs. The solution to the first stage defines the deadweight costs of the tax system, $C(G) = C(R^{I*}(G)) + C(R^{O*}(G))$. In the second stage, the government chooses the optimal size

tax; that is clearly inconsistent with our findings as we see states relying more on the income tax following the adoption of withholding. Formulating $a\tau^I$ as the effective tax rate also implies that taxpayers can evade costlessly; this too strikes us as being unreasonable (e.g., Saez, Slemrod, and Giertz, 2012, and Thomas, 2015).

¹⁵The independence assumptions are relevant because if an improvement in collections of the personal income tax had positive spillover effects on other taxes, revenue from other taxes would also rise holding tax rates constant. Our empirical estimates confirm the validity of the independence assumption; they do not reveal any spillovers from the personal income tax to other taxes. See Tables 2 and A.11 (Online Appendix) and the associated discussion in Bagchi and Dušek (2021) for further details.

¹⁶Incorporating all elements of the costs would make the notation much more complex without additional insights.

¹⁷Although the direct choice variables are the tax rates instead of revenues, it is analytically much more convenient to solve the model for revenues in which case the tax rates then drop out according to $\tau^I = \frac{R^I}{aY^I}$ and $\tau^O = \frac{R^O}{Y^O}$.

of government that maximizes $\gamma B(G) - C(G)$.

The solution to the first stage gives the familiar result that the marginal deadweight costs per dollar of revenue must be equalized across both taxes. It is straightforward to verify that

$$C(G) = \frac{1}{2} \frac{\delta \epsilon}{a \delta Y^I + \epsilon Y^O} G^2 = \frac{1}{2} c G^2 \quad (6)$$

$$R^{I*}(G) = \frac{a \delta Y^I}{a \delta Y^I + \epsilon Y^O} G = \frac{a Y^I}{\epsilon} c G = s^I G \quad (7)$$

$$R^{O*}(G) = \frac{\epsilon Y^O}{a \delta Y^I + \epsilon Y^O} G = \frac{Y^O}{\delta} c G = s^O G \quad (8)$$

$$\frac{R^{I*}}{R^{O*}} = \frac{\delta}{\epsilon} \frac{a Y^I}{Y^O} \quad (9)$$

Both tax sources increase proportionally with total taxes and are inversely related to their deadweight cost parameters. The advantage of this formulation is that the terms $\frac{a Y^I}{\epsilon} c$ and $\frac{Y^O}{\delta} c$ denote the shares s^I and s^O of the personal income tax and other taxes respectively, which are observable in the data.

The solution to the second stage is given by equating the marginal benefit of spending to marginal deadweight costs, and implicitly defines the demand for government revenues as a function of γ and the deadweight cost parameter c :

$$\gamma B' = c G \quad (10)$$

Withholding is represented in the model as an increase in a . The comparative statics on a show the percentage equilibrium changes in revenues due to withholding:

$$\frac{dG}{G} = \left(\frac{Y^I}{\epsilon} \frac{c^2}{c - \gamma B''} \right) da > 0 \quad (11)$$

$$\frac{dR^I}{R^I} = \frac{da}{a} + \frac{d\tau^I/da}{\tau^I} da = \left(\frac{ds^I/da}{s^I} + \frac{dG/da}{G} \right) da = \left(\frac{1}{a} + \frac{c Y^I}{\epsilon} \frac{\gamma B''}{c - \gamma B''} \right) da > 0 \quad (12)$$

$$\frac{dR^O}{R^O} = \frac{d\tau^O/da}{\tau^O} da = \left(\frac{ds^O/da}{s^O} + \frac{dG/da}{G} \right) da = \left(\frac{c Y^I}{\epsilon} \frac{\gamma B''}{c - \gamma B''} \right) da < 0 \quad (13)$$

Equation (11) shows that spending should increase, because it is socially less costly to raise an additional dollar of revenue. Equation (12) first decomposes the change in income tax revenue into the mechanical and political effects. The mechanical effect denotes the increase in income tax revenue driven by the improvement in compliance, equaling da/a percent. Additionally, as the current tax rate is no longer optimal, the government responds by adjusting the tax rate downward. We

refer to this as the political effect and it reduces income tax revenue by $d\tau^I/\tau^I$ percent.

The political effect reflects the sensitivity of demand for public spending to the marginal deadweight cost. The model predicts that $d\tau^I/\tau^I \in (-da/a, 0)$, i.e., the tax rate should decrease but by less than the mechanical effect, such that personal income tax revenues still increase.¹⁸ This is an opposite prediction to Friedman’s conjecture that withholding caused further growth in the personal income tax. Friedman’s conjecture, translated into the language of the model, implies $d\tau^I/\tau^I > 0$, i.e. the political effect is so large that the tax rate would rise and revenues would increase “on top of” the mechanical effect.¹⁹

For other taxes, the mechanical effect is absent, and the political effect is negative. As the marginal deadweight cost of the personal income tax decreases, it is optimal to reduce the revenue from other taxes in order to equalize the marginal deadweight costs across the two taxes (equation 9). For that reason, the tax rate τ^O should be decreased as well.²⁰

The analysis has to be modified if the change in a is endogenous, which is likely to be true here as states adopted withholding voluntarily and for a reason. Withholding is a technology that generates more revenue, so it is natural to expect adoption when governments need more revenue. In that case, we would observe an improvement in tax collection ($da > 0$) following the introduction of withholding around the same time when there is also an increase in the demand for government revenue ($d\gamma > 0$).

When adoption is driven by an increase in demand for spending, the simultaneous increase in

¹⁸The political effect thus mitigates the mechanical effect; the government “returns” a part of the increase in revenue to the taxpayers by cutting the tax rate.

¹⁹Other responses that lie outside of the predicted interval also have interesting interpretations: If $d\tau^I/\tau^I = -da/a$, i.e., the government would reduce the tax rate such that it fully offsets the extra revenue brought by the mechanical effect. This would imply that the revenues are irresponsive to an improvement in tax collections. If $d\tau^I/\tau^I = 0$, i.e., the government merely keeps the mechanical increase in revenue and there is no political effect.

²⁰The second terms in equations (12) and 13 present an alternative decomposition into substitution and scale effects. A reduction in the marginal deadweight costs of the personal income tax implies that for a given amount of total revenue, personal income tax revenue should increase and revenue from other taxes decrease in order to equalize marginal deadweight costs across tax sources. This is the substitution effect, increasing the share of the income tax ($\frac{ds^I}{s^I} \frac{da}{a}$). In addition, because the marginal deadweight cost of the overall tax system has declined, there should be a scale effect. Revenue from personal income and other taxes should increase proportionately by the same percentage with the increase in spending ($\frac{dG}{G} \frac{da}{a}$). Thus for the personal income tax, the substitution and scale effects have the same sign. (Our use of the terms, “substitution effect” and “scale effect”, is similar to usage in labor economics when analyzing the change in demand for a factor of production in response to a price change (e.g. Cahuc and Zylberberg, 2004).) The substitution effect and scale effect for other taxes however have the opposite sign, with the substitution effect ($\frac{ds^O}{s^O} \frac{da}{a}$) being negative and the scale effect ($\frac{dG}{G} \frac{da}{a}$) being positive. The fact that the revenue from other taxes should decrease can be interpreted as the substitution effect dominating the scale effect; as total spending increases, the marginal benefit of spending decreases, which means that the marginal deadweight costs of both the personal income tax and other taxes must decrease. Because there is no improvement in tax collections for other taxes, revenue from other taxes must decrease in order to reduce the marginal deadweight costs.

a and γ has implications for the interpretation of the *observed* changes in tax revenues after withholding. In addition to the mechanical and political effects, tax revenues will change in response to the demand shock. The demand shock is now explicitly added as the right-most term in the equations below:

$$\frac{dR^I}{R^I} = \frac{dR^I/da}{R^I}da + \frac{dR^I/d\gamma}{R^I}d\gamma = \left(\frac{da}{a} + \frac{d\tau^I/da}{\tau^I}da \right) + \frac{dG/d\gamma}{G}d\gamma > 0 \quad (14)$$

$$\frac{dR^O}{R^O} = \frac{dR^O/da}{R^O}da + \frac{dR^O/d\gamma}{R^O}d\gamma = \frac{d\tau^O/da}{\tau^O}da + \frac{dG/d\gamma}{G}d\gamma \leq 0 \quad (15)$$

$$\text{where } \frac{dG/d\gamma}{G} = \frac{c}{\gamma(c - \gamma B'')} > 0 \quad (16)$$

Personal income tax revenues increase by more than what the model predicts when withholding is exogenous (compare eqs. (14) and (12)), and the personal income tax rate may decrease or increase. Similarly, other taxes do not decrease by as much, and they may even increase (compare eqs. (15) and (13)). Only the mechanical and political effects represent the causal relationship from withholding to revenues; the response to the demand shock, $\frac{dG/d\gamma}{G}d\gamma$, represents the reverse causality from the need for higher spending to the introduction of withholding.

Although the model is flexible and allows for a range of possibilities, nonetheless it imposes restrictions that allow us to interpret our observed changes in tax revenues. First, the demand shock is unambiguously present if revenues from other taxes (and thus other tax rates) increase. Second, if only the demand shock were present, but the choice of revenues did not adjust to the introduction of withholding, then revenues from both tax sources would increase by the same percentage. Third, a relative increase in personal income tax revenues is consistent with the optimal adjustments to withholding as implied by the mechanical and political effects.²¹

In the empirical section, we estimate the revenue responses to the adoption of withholding and interpret them through the lens of this model. Specifically, we assess whether the observed changes in revenues are consistent or not with the model's predicted adjustments in revenue choices due to an improvement in the ability of governments to collect taxes. Furthermore, we assess whether the observed changes are consistent with either an exogenous or an endogenous adoption of with-

²¹Likewise, the interpretation can be phrased in terms of the substitution and scale effects: If the scale effects were solely driving the changes in revenues, both taxes would increase by an equal percentage. The substitution effect is unambiguously present if the increase in income tax revenue exceeds the increase in revenue from other taxes. Observing a decline in other taxes, on the other hand, would be consistent both with the absence of the demand shock as well as with a small positive demand shock that is dominated by the sum of the substitution and scale effects.

holding, i.e. the absence or presence of a positive demand shock driving the adoption decision.

4 Empirical strategy

In this section we derive an econometric model that directly reflects the structure of the theoretical model. The subscripts i and t denote state and year respectively. All financial variables (revenues and income) are in real per capita terms.

The following equations capture the underlying demand for tax revenues by each source, that is, revenues from the personal income tax, R_{it}^I , and from other taxes, R_{it}^O , that the government desires to raise:

$$R_{it}^I = (1 + \delta^I D) r^I(y_{it}, X_{it}) e^{u_{it}^I} \quad (17)$$

$$R_{it}^O = (1 + \delta^O D) r^O(y_{it}, X_{it}) e^{u_{it}^O} \quad (18)$$

The dummy variable, D_{it} , indicates whether a state that already has a personal income tax in place makes use of withholding or not. The total effects of withholding – the overall percentage changes in revenues – are captured by parameters δ^I and δ^O . The functions $r^I(\cdot)$ and $r^O(\cdot)$ are empirical counterparts of the parameter γ in equation (5). They are a function of the income per capita, y_{it} , and socio-economic variables, X_{it} , which plausibly affect the technology of collection and the demand for tax revenue. u_{it}^I and u_{it}^O are error terms.

To estimate δ 's, we assume a log-linear functional form for r^I and r^O :

$$\log r^I = \gamma_X^I \log X_{it} + \gamma_y^I \log y_{it} \quad (19)$$

and

$$\log r^O = \gamma_X^O \log X_{it} + \gamma_y^O \log y_{it} \quad (20)$$

Taking logs of (17) and (18) and substituting (19) and (20), we obtain the estimating equations:

$$\log R_{it}^I = \log(1 + \delta^I D) + \gamma_X^I \log X_{it} + \gamma_y^I \log y_{it} + u_{it}^I \quad (21)$$

$$\log R_{it}^O = \log(1 + \delta^O D) + \gamma_X^O \log X_{it} + \gamma_y^O \log y_{it} + u_{it}^O \quad (22)$$

Equations (21) and (22) are estimated by OLS. In all regressions, state and year dummies are

included to capture unobservable fixed effects that are constant within a state over time or across states within a year. In a subset of specifications, we include state-specific linear time trends thus allowing tax revenues to evolve separately in each state, in order to account for the growth in state tax revenues over the sample period, as well as changes in the structure of the economy such as a decline in self-employment and increasing formalization. Following Bertrand, Duflo, and Mullainathan (2004), we cluster standard errors by state all throughout.²²

This exercise produces the estimates of the total effects, δ^I and δ^O . To disentangle δ^I further, we first obtain an estimate of α , the mechanical increase in revenues holding tax rates constant. This was the main focus of our earlier paper (Bagchi and Dušek, 2021) and is replicated here only briefly.²³ The equation that describes the personal income tax collection technology is defined as:

$$\log R_{it}^T = \log(1 + \alpha)D_{it} + a_\tau \log \tau_{it} + a_X \log X_{it} + (1 + a_y) \log y_{it} + u_{it}^T \quad (23)$$

With the estimates $\hat{\alpha}$ and $\hat{\delta}^I$ at hand, we can compare them to test whether the overall increase in personal income tax revenues (δ^I : due to the sum of the mechanical effect, political effect, and the demand shock) exceeds the mechanical effect (α). Likewise, it is useful to test for the equality between $\hat{\delta}^I$ and $\hat{\delta}^O$. In addition to estimating the effects of withholding on revenues, we estimate its effects on the personal income tax and other tax rates. The estimation is analogous to equations (21) and (22); the revenues on the left-hand side are merely replaced by the corresponding tax rates. The following table summarizes our predictions for the various scenarios in which (A) revenues either (1) respond to the introduction of withholding or (2) they don't and (B) where the adoption of withholding is either (1) exogenous or (2) driven by a positive demand shock:

²²Although clustering by state addresses a possible correlation of errors for a state over time, another concern may be that errors are correlated across states. This may stem from states competing with each other in choosing aspects of their tax systems or due to a combination of real and reporting responses by taxpayers. However, we do not find any evidence that spatial correlation across states affects our estimates of the effect of withholding on the personal income tax. Using the Stata command, `xsmle` (Belotti, Hughes, and Mortari, 2017) that allows us to model spatial correlation with panel data in alternative ways (either as a spatial lag in the dependent variable and/or a spatial lag in the independent variables and/or a spatial lag in the error term), we obtain results that are practically identical to those obtained using OLS. Also, in a similar vein, our results are robust to clustering standard errors at the region level. However, there are only 4 Census regions and it is well-known that clustering with less than 30 groups can lead to wildly misleading standard errors (e.g., Bertrand et al. (2004) and Cameron and Miller (2015)). Given that there are 29 states in our sample, which is close to the rule of thumb of 30 groups, by clustering at the state level we avoid this critique. Our additional results involving the use of `xsmle` or clustering at the region level are available on request.

²³Equation 23 is derived analogously to the demand equation.

Scenario	Do revenue choices causally respond to withholding?	Is the adoption of withholding exogenous or driven by a positive demand shock?	Prediction on personal income tax and other tax revenues	Prediction on income tax rate and other tax rates
1	No	Exogenous	No change ($\widehat{\delta^I} = \widehat{\delta^O} = 0$)	Income tax rate: Decline by the mechanical effect; Other tax rate: No change
2	No	Driven by a positive demand shock	Both increase by the same percentage, the exact magnitude of the increase depending on the size of the demand shock ($\widehat{\delta^I} = \widehat{\delta^O} > 0$ & $\widehat{\delta^I} \leq \widehat{\alpha}$)	Income tax rate: Either decrease or increase depending on the size of the demand shock; Other tax rate: Increase
3	Yes	Exogenous	Personal income tax revenues: Increase, but by less than the mechanical effect; Other tax revenues: Decrease ($\widehat{\delta^I} > 0$, $\widehat{\delta^I} < \widehat{\alpha}$, and $\widehat{\delta^O} < 0$)	Both tax rates: Decrease
4	Yes	Driven by a positive demand shock	Personal income tax revenues: Increase by less or more than the mechanical effect depending on the size of the demand shock; Other tax revenues: Increase ²⁴ ($\widehat{\delta^I} > 0$, $\widehat{\delta^I} > \widehat{\delta^O}$, and $\widehat{\delta^I} > \widehat{\alpha}$)	Personal income tax rate: Either decrease or increase depending on the size of the demand shock; Other tax rate: Increase

²⁴ $\widehat{\delta^O}$ could be more than or less than zero depending on how large the demand shock is relative to the size of the mechanical effect driving higher compliance with the personal income tax and the resulting adjustment by governments to the revenue mix. $\delta^O > 0$ unambiguously indicates a positive demand shock whereas $\delta^O < 0$ indicates either the

Before proceeding with our estimation and presenting the regression results obtained using OLS, we acknowledge the growing literature that has expressed concerns about the validity of the conventional two-way fixed effects estimator. Although we consider those concerns, we find them to have less bite in our context as our estimates suggest the lack of dynamic treatment effects as well as the lack of heterogeneous treatment effects (refer, Sec. A of the Online Appendix). Moreover, as the example in Section 6.3 for the state of California suggests, the institutional context largely rules out anticipation effects.

5 Data

Our dataset is an annual panel of the 48 contiguous U.S. states from 1944 till 2000. Alaska and Hawaii are excluded because fiscal data for those states are available only from 1959 and by that time, both states had already implemented withholding.

The information on when each state introduced the income tax and implemented withholding comes from Penniman (1980). Data on personal income tax rates and brackets, along with tax base variables (e.g., personal exemption, standard deduction, etc.), are based on Bakija (2017). The Michigan World Tax Database and The Book of the States, a publication of The Council of State Governments,²⁵ are the sources of data on rates for the corporate income, sales, and excise taxes, with any differences between the two sources reconciled through an extensive search of newspaper archives. The data on state government tax revenues (by each tax source), general expenditures (broken down into six major categories), and debt are taken from “Compendium of State Government Finances”, an annual publication of the U.S. Census Bureau. Government revenue and expenditure variables were converted into real 1982 dollars by the “state and local government expenditure deflator”.²⁶

A number of socio-economic controls are included in the regressions. Factors likely to influence the demand for tax revenue and the technology of tax collection are the size of the state

absence of a demand shock or a demand shock that is sufficiently small.

²⁵The publication describes legislative changes occurring in each calendar year, which makes it possible to fill in the tax rates for those years that are not reported in biennial tables.

²⁶Both indices are published in the Economic Report of the President.

population,²⁷ personal income, share of farm income,²⁸ and the share of interest and dividend income in personal income, data for which come from the Bureau of Economic Analysis, Regional Accounts Data. The personal income variables were converted into real 1982 dollars by the consumer price index. Also included are the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas, the data for which were obtained from Census enumerations.²⁹ Following literature that examines the causes for government growth, we include a number of variables likely to affect the demand for tax revenue: female labor force participation (Kau and Rubin, 1981), the share of income going to the top 1 percent of the population (Peltzman, 1980), and the average vote share of Democrats in all state-wide races (Reed, 2006). Data on these variables were obtained from Census enumerations, the State-Level Income Inequality Data developed in Frank (2009, 2014), and a dataset originating in the work of Ansolabehere and Snyder (2002) respectively.³⁰ Table 2 presents summary statistics for all variables used in the regressions.

[Table 2 about here.]

6 Results

Before presenting our regression results, we present a figure, Figure 2, that conveys the essence of our results. The sub-figure in the top left corner plots personal income tax revenues before and after the adoption of withholding. The vertical axis shows the real per capita personal income tax revenues, averaged across the 29 states which had an income tax in 1944 at the start of our sample period. All state years on the horizontal axis are normalized so that year zero is the fiscal year in which a state implemented withholding. We can see from this figure that real per capita revenues from the personal income tax grow steadily until they reach 60 dollars prior to the adoption of withholding and they then jump sharply to about 85 dollars in the year of adoption. Although the figure has to be read with caution as no other factor affecting income tax revenues are controlled

²⁷We examine the responses of population to the introduction of withholding but do not find any significant responses. That result is in line with findings in Cassidy, Dincecco, and Troiano (2024) who report no significant outmigration for states that adopted the personal income tax prior to World War II in contrast to outmigration of middle and high-earning households in states that introduced the personal income tax after World War II. Given that our focus throughout the paper (with the exception of Subsection 7.3) is on states that had an income tax prior to World War II, our results, combined with the lack of significant outmigration reported by Cassidy et al. (2024), reduce endogeneity concerns around including population as a socio-economic control. Furthermore when we estimate weighted regressions, with state population in 1944 as weights, the results we obtain are identical to those in our baseline specification.

²⁸The farm income series is very volatile and is negative for some states in some years. Because this variable is meant to capture the long-term agricultural vs. industrial character of the state rather than short-term shocks to agricultural incomes, we use three-year moving averages of farm income in the analysis.

²⁹As Census data are decennial, observations for the intervening years were generated by linear interpolation.

³⁰We are indebted to Jim Snyder for sharing an updated and extended version of this dataset with us.

for, it strongly suggests that withholding had a sizable impact on personal income tax revenues.

[Figure 2 about here.]

The second sub-figure in the top right corner shows that there was an increase in other tax revenues as well. While we analyze trends in other taxes formally later on, this figure does suggest that revenues from other taxes did not go down to offset the increase in taxes resulting from improved collection of the personal income tax; in fact, as we see subsequently, they also went up, albeit modestly, following the introduction of withholding.

In contrast to these sharp increases in tax revenues, the third sub-figure in the bottom left corner shows that there were no sharp increases in state expenditures around the time of adoption of withholding. The final sub-figure in the bottom right corner offers a possible explanation for this apparent puzzle that tax revenues went up following the adoption of withholding without a corresponding increase in expenditures and it involves state debt. Although the ratio of state debt to revenue had increased from around 34 percent to 53 percent during the 8 years prior to the adoption of withholding, debt levels plateau after the adoption of withholding. The ratio of debt to revenues declines by about 3 percent in the two years following the adoption of withholding and never exceeds the level we see in the year right prior to the adoption of withholding. Although we examine these issues more formally and control for many possible determinants of state revenues and expenses, the basic message conveyed by this figure holds up in the subsequent regression analysis.

6.1 Effects of introducing withholding on taxes

We first examine the effects of the introduction of withholding on the state personal income tax in Table 3. Column (1) presents the most parsimonious specification with no controls other than state and year fixed effects with column (2) additionally including state-specific linear time trends. Columns (3) and (4) exclude state-specific time trends but they include controls likely to have an effect on the ease of collection as well as on the demand for tax revenue: variables such as the percent of the elderly population and the degree of urbanicity. The additional controls included in column (4) are those that are likely to primarily affect the demand for tax revenues, viz. female labor force participation, a measure of income inequality, and partisan preferences of the population

as captured by the average Democratic vote share in all state-wide races. Column (5) reintroduces state-specific time trends in addition to these controls. Finally, column (6) adds controls that are potentially endogenous – the top and bottom personal income tax rates. The coefficients on withholding in Columns (1) through (5) have the interpretation of the “overall effect” of withholding – the percentage change in revenue due to the mechanical and the political effects combined plus any unobservable demand shock, if present – whereas the coefficient on withholding in column (6) captures solely the mechanical effect – the increase in revenues holding tax rates constant.³¹

[Table 3 about here.]

The coefficients on withholding in the first row of Table 3 suggest that the adoption of withholding had a significant impact on personal income tax revenues. The coefficients in columns (1) through (5) are clustered in a range between 28.4–39.2 percent and are each significant at the 1% level (p-value < 0.001).³² The most conservative estimate obtained in column (5) is 0.284 and it means that state governments began collecting an additional 28.4 percent in personal income taxes following the adoption of withholding. Lastly, the estimate of 0.217 in column (6) measures the mechanical effect, i.e. the increase in personal income tax revenues driven solely by higher compliance.³³

The comparison between the overall ($\hat{\delta}^I$) and the mechanical ($\hat{\alpha}$) effects is important for the interpretation of findings. We find that the overall effect (0.284, column (5)) exceeds the mechanical effect (0.217, column (6)) by over 30 percent, and a formal test shows that the two coefficients are different from each other at the 1% level of significance. These findings suggest that only a part of the post-withholding increase in personal income tax revenues can be accounted for by

³¹As we establish in Bagchi and Dušek (2021), there is no change in various measures of the tax base around the time of adoption of withholding. Accordingly, we do not introduce those measures of the tax base as controls in col. (6).

³²Because Oregon and Delaware adopted withholding in 1948 and 1949 respectively, we do not have 8 years of data prior to their adoption. Accordingly, the underlying data used in the estimation are unbalanced. For robustness, we provide “balanced panel” versions of our estimates in the Online Appendix. Those estimates in Table A.1 guard against changes in group composition driving our results. We also find our results robust to dropping the period immediately following the federal introduction of withholding in 1944 and have our sample period start in 1950 or 1952 instead for instance.

³³Note that the reported estimate of the mechanical effect is not the same as the most preferred estimate (0.290, see Table 1, Col. 4) of the same effect in Bagchi and Dušek (2021). The key reason for the difference in the two estimates are the inclusion of state-specific trends here and their exclusion in Bagchi and Dušek (2021) upon the explicit request of the journal editor. Specifications with state-specific trends are in general less restrictive and, in our setting, produce more conservative estimates; for those reasons we prefer to present them here. A secondary reason for the difference in estimates across papers is the inclusion of three additional socio-economic variables that capture the underlying demand for government revenue – variables that were absent in Bagchi and Dušek (2021) where the focus was strictly on the effects of introducing withholding on income tax compliance.

the mechanical effect and that the subsequent political adjustment as well as a demand shock contributed additionally to the increase in collections.

In Table 4, we examine the effects of the adoption of withholding on other sources of tax revenue – both in the aggregate – and then decomposed by source, such as the corporate income tax, the sales tax (including both the general sales tax and the excise tax), and all other sources (e.g., license taxes and state property taxes). The manner in which the regressions are presented mimic those in Table 3, with progressively more controls included as we move from left to right and the final specification including state-specific linear time trends and all socio-economic controls.

[Table 4 about here.]

In contrast to the robust increase in personal income tax revenues we observe, results for the other taxes are more mixed. Referring back to column (5) of Table 4 which generally leads to the most conservative estimates, we find suggestive evidence that all other taxes, corporate income taxes, and sales taxes increased by 1.3, 8.4, and 1.8 percent respectively following the adoption of withholding although none of these estimates are statistically significant. Estimates for the catch-all category, “other taxes”, which includes property, license, death and gift, poll, and other miscellaneous taxes bounce around the most but here too the most conservative estimates in column (5) suggests an increase of 1.4 percent, although the estimated change is, again, statistically insignificant. Thus, the estimated percentage increase in corporate income taxes and sales taxes – both general and selective – as well as other taxes is lower than the estimated increase in personal income taxes. This statement is also true when we consider taxes in levels: One year before the adoption of withholding, personal income taxes were \$59 per capita and so the 28.4 percentage increase (from column (5) of Table 3) implies an absolute increase of \$16.69. On the other hand, corporate income tax revenues, sales tax revenues, and other revenues were, on average, \$27, \$212, and \$98 per capita in the year before adoption and the estimated percentage increases (from column (5) of Table 4) represent an absolute increase of \$2.27, \$3.91, and \$1.37 respectively.

To recap the results of this sub-section, we find a large increase in personal income tax revenues following the adoption of withholding by about 28 percent with weak evidence of a modest increase in revenues from the corporate income tax, the sales tax, and other miscellaneous taxes. The differential increase in personal income tax revenues *vis-a-vis* other taxes suggests that the share of the personal income tax in total revenues would have increased. This point is confirmed

by the regressions presented in Table 5 (and the visual evidence in Figure A.1) in which we directly examine the impact of introducing withholding on the share of taxes collected from the personal income tax. In the most conservative specification reported in Table 5 that includes state-specific trends and all control variables, the share of taxes collected from the personal income tax increases by 0.238 or 23.8 percent.

[Table 5 about here.]

Relating these findings to the testable predictions from Sections 3 and 4, we can clearly reject the null hypothesis that revenues are irresponsive to withholding. We observe increases in revenues (inconsistent with Scenario 1 that predicts no increase in taxes) and an increase in the share of revenues from the personal income tax (inconsistent with Scenario 2 that predicts an equal percentage increase in all taxes). Governments clearly responded to the improvement in the collections of the personal income tax by shifting the composition of revenues towards that tax and increasing personal income tax revenues. These effects are consistent with the model's prediction for both exogenous adoption (Scenario 3) and adoption driven by a demand shock (Scenario 4). However, what rules out Scenario 3 is that under exogenous adoption, revenues from other taxes should decline and the overall increase in personal income tax revenues should be smaller than the mechanical effect, but we see no evidence for either prediction in the data. Instead we observe small, although insignificant, *increases* in other tax revenues, and moreover, the overall effect of withholding on personal income tax revenues exceeds the mechanical effect. Hence, our findings are most in line with Scenario 4, viz., the predicted adjustment in revenues from both tax sources (as collecting the personal income tax becomes easier) *and* the presence of a demand shock that likely drove the adoption of withholding in the first place.

6.2 Checking for pre-existing trends and additional robustness checks

A fundamental concern in any difference-in-differences analysis is the possibility that the observed mean effects are being driven by differences in pre-existing trends and that these trends may be biasing our estimates. We address that important concern in two ways.

First, to check for possible unobservable trends and to obtain additional insights about the speed of response of tax revenues to withholding, we estimate the same equation as in Table 3

but replace the withholding dummy with dummy variables representing the year of adoption, the 1st, 2nd, etc. up to 7th year after the adoption of withholding, and a dummy variable equal to 1 for 8 and more years after adoption. We also include dummy variables for the 2nd, 3rd, etc. up to 7 and another dummy variable for 8 (or more) years before the adoption of withholding.³⁴ For expositional clarity, only the coefficients on the dummy variables from selected equations and their 95% confidence intervals are plotted against the time axis in Figure 3.³⁵

[Figure 3 about here.]

Figure 3 shows that there was no unobservable trend in personal income tax revenues before the adoption of withholding given that the coefficients are fairly precisely estimated and yet not statistically different from zero. The effect of withholding also appears immediately and is persistent: income tax revenues rise by over 30 percent in the adoption year and the effect sizes in the following years are not statistically different from that in the adoption year. A (statistically insignificant) drop that is visible in the second and third post-adoption years is due to a particular institutional feature. In most states, taxpayers had to remit the newly withheld taxes as well as the last year's tax liability, generating a one-time fiscal windfall on top of the real improvement in tax collections. This transitory shock is, on average, spread across the adoption year and the first post-adoption year because the date of adoption did not typically coincide with the beginning of the fiscal year.³⁶ The immediacy of the effect is noteworthy and the sharp jumps make it unlikely that the increased collections were the product of other changes afoot when states adopted withholding. Once this transitional dip in the second and third post-adoption years dissipates, the effect gradually grows back to about 35 percent over the next five years.

Figure 4 plots an analogous picture for all other taxes; specifically, the coefficients on the pre- and post-withholding dummy variables for all state taxes but the personal income tax. After controlling for other determinants, these taxes were actually on a downward trend before the adoption of withholding. The trend reverses exactly when withholding is adopted – these taxes jump up by about 1.6 percent and continue to be statistically distinguishable from zero in the post-withholding period. The switch in the trend for other taxes, and especially their discrete jump in the adoption year, indicates that a demand shock may have motivated the adoption of withholding.

³⁴The dummy variable representing the last year before adoption is dropped so that it serves as the baseline.

³⁵Complete results from the regressions are available upon request.

³⁶To reassure the reader of the robustness of our results, we present Robustness Check 2 in Table 6 in which we exclude the first two years following the adoption of withholding.

[Figure 4 about here.]

We take a second approach to address the question of whether pre-existing trends bias the coefficients obtained on the withholding dummy in Table 3. We do so by implementing a permutation test in which we take the set of states adopting withholding, and randomly assign to each state a year of adoption.³⁷ We then estimate our most conservative specification with state-specific linear time trends and all time-varying controls and record the coefficient on the withholding dummy. We repeat this exercise 5,000 times, and investigate the share of placebo coefficients that are larger than the baseline estimates. The results are presented graphically in Figure 5 and they show that our estimated impact of withholding is much larger than the placebo estimates and underlying trends in tax collection are unlikely to confound our estimate.³⁸

[Figure 5 about here.]

Finally, in Table 6, we present a battery of robustness checks that address the pre-existing trends and other plausible concerns. A complete description of these checks is relegated to Online Appendix C. In brief, (i) the checks find a weakly negative trend in the personal income tax revenues and no pre-trends in other tax revenues. Also, the estimates of introducing withholding on personal income tax revenues and the share of revenues from this tax are robust to (ii) omitting the transitory period after adoption (iii) including state-specific quadratic time trends, (iv) including Census region-by-year fixed effects, (v) including Census division-by-year fixed effects, (vi) estimating weighted regressions, with state populations (either current levels or using population as of 1946 serving as weights), (vii) using a random effects specification, (viii) controlling for the presence of information sharing agreements between the states and the federal government (Troiano, 2018), and (ix) splitting the sample of states based on the existence (or lack thereof) of a strict balanced budget requirement.

[Table 6 about here.]

³⁷The adoption years are drawn from a normal distribution which appears to fit the adoption pattern fairly well.

³⁸We undertake identical permutation tests for the share of taxes from the personal income tax and the revenues from all other taxes but the personal income tax. Our results are practically unchanged with the former variable and the permutation test suggests that the estimated impact of withholding is much larger than all of the placebo estimates. The permutation tests for revenues from all other taxes paints a more mixed picture but here too our estimated effect is larger than about 70 percent of the placebo estimates. These additional results are available on request.

6.3 The lack of anticipation effects – The case of California

An assumption implicit in the difference-in-differences framework is the lack of anticipation effects (Sun and Abraham, 2021). We illustrate the validity of this assumption in our context offering the example of California’s adoption of withholding (and three additional case studies in Online Appendix B). Of the states that had an income tax at the end of World War II, California was the second last state to introduce withholding in 1971. Although attempts had been made by Governor Edmund G. (Pat) Brown, a Democrat, to introduce withholding as early as 1963, he failed to persuade a Democratic-controlled General Assembly to agree with his proposed introduction of withholding.³⁹ Even the subsequent enactment of withholding eight years later in 1971 resulted from an unusual confluence of events. Ronald Reagan, who had first been elected Governor of the state in 1966, had been adamantly opposed to withholding and is reported to have said that: “My feet are in concrete on my opposition to withholding.”⁴⁰ Holding true to those views, Governor Reagan was reconciled to the legislature rejecting his tax reform proposals during his first term in office if a compromise meant agreeing to the introduction of withholding⁴¹ as he viewed withholding as “an opiate, a narcotic” that would make it easy for government to raise taxes without the people noticing it.⁴² Eventually though during his re-election campaign in 1970, Governor Reagan appeared willing to adopt withholding,⁴³ and in 1970 – right after re-election – he came close to signing a comprehensive tax reform bill that introduced withholding, only to have that bill be defeated in the state Senate by *one* vote.⁴⁴ A year later, finally, as part of a deal with the Democratic majority on reduction in property tax burdens for homeowners⁴⁵ and legislative reapportionment, Governor Reagan signed legislation introducing withholding in California.⁴⁶ A tax forgiveness proposal – forgiving 20 percent of taxpayers’ 1971 tax liability – similar in spirit to what had been used during the introduction of federal income tax withholding was crucial to securing its passage in the legislature.⁴⁷ Finally note that even as withholding was signed into law on 8th December

³⁹“Adversaries gird for battle,” p. 6, Daily News-Post, 21 Dec 1970

⁴⁰“Income Tax Withhold Bill Signed,” p. 16, Oakland Tribune, 08 Dec 1971

⁴¹“Reagan Still Opposes Tax Withholding Plans,” p. 41, The Los Angeles Times, 04 June 1969

⁴²“Governor blisters Assembly Democrats over ‘Fun, Games’,” p. A8, The Sacramento Bee, 03 June 1967.

⁴³“Withholding, sales tax hike, property tax relief package,” p. 1, Ukiah Daily Journal, 04 Feb 1970

⁴⁴“Reagan Softens Concrete Stand on Withholding,” p. 6, The Californian, 08 Dec 1971

⁴⁵“Reapportion bid a time consumer,” p. 4, News-Pilot, 04 Dec 1970

⁴⁶“Solons Face Another Tough Year,” p. 13, Chico Enterprise-Record, 31 Dec 1971

⁴⁷“Withholding Forgiveness Eases Jolt,” p. 1, Petaluma Argus-Courier, 09 Dec 1971

1971, it came into effect on 1st January, 1972, less than four weeks later,⁴⁸ pointing to the lack of anticipation effects of this reform. The case of California illustrates well how the timing of adoption was influenced by idiosyncratic circumstances and the quirks in personalities of the political leaders within the state rather than its specific fiscal needs.

6.4 Examination of tax rates

Examining changes to tax rates set by states around the period when withholding is adopted is a useful exercise as it provides additional evidence for discriminating between the alternative scenarios. Thus, if the introduction of withholding – and the subsequent improvement to tax compliance – were accompanied with hikes to tax rates, it would suggest that governments that introduced withholding had a pressing need for higher revenues and in the course of responding to this need they adopted withholding *and* increased tax rates. On the other hand, failing to find significant increases to tax rates would suggest that the adoption was perhaps not motivated by a need to raise more revenue. We present the results of an examination of tax rates in Table 7, with Panels A and B corresponding to the top and bottom personal income tax rates, Panel C examining the corporate income tax rate, Panel D the general sales tax rate, and Panel E the excise tax rate on gasoline.⁴⁹

[Table 7 about here.]

Table 7 suggests a large and statistically significant increase in the top personal income tax rate and the corporate income tax rate of about 15 percent. There appears to be a more modest increase in the bottom personal income tax rate of about 6 percent, although that increase is not statistically significant. Finally, there are no effects of the introduction of withholding on either the general sales tax rate or the excise tax rate on gasoline, with coefficients changing sign based on the specification and generally small in magnitude. The increase in the top personal income tax rate and the corporate income tax rate – notwithstanding the higher revenues that states were already collecting through improved compliance with the personal income tax – is consistent

⁴⁸“Curb Sought on Tax Hikes,” p. 3, Times-Advocate, 09 Dec 1971

⁴⁹The reason for examining the tax rate on gasoline is that states raise significantly more in revenues by taxing gasoline as compared to taxing other commodities such as alcohol and tobacco. See, <https://www.urban.org/policy-centers/cross-center-initiatives/state-and-local-finance-initiative/state-and-local-backgrounders/state-and-local-revenues#state>, for recent figures on excise tax collections from motor fuel, alcohol, and tobacco. Additionally, each of the 29 states included in our analysis imposes an excise tax on gasoline making it easily amenable to analysis.

with a narrative in which states had a pressing need for revenues and enacted a host of measures to meet that need. The increase in the top personal and corporate income tax rates following the introduction of withholding is a robust result that holds up to numerous checks, including introducing state-specific quadratic time trends, region-by-year fixed effects, division-by-year fixed effects, and weighted regressions with state population used as weights (See Table A.2 in the Online Appendix). Furthermore, examining the dynamics of the rate increases as we do in Figure A.2 offers additional evidence consistent with a demand shock: Both the top personal income tax and corporate income tax rates increase right around the time that withholding was adopted.

6.5 Response of governments to the additional revenue

The results of our empirical analysis confirm increases in tax revenues by about 5 percent in the aggregate following the adoption of withholding, resulting both from higher compliance with the personal income tax, as well as increases to the top personal income and the corporate income tax rates. A reasonable question to ask is: What did state governments do with the money?

To shed some light on that question, we first estimate the effect of withholding on state general expenditures and subsequently on seven expenditure categories into which general expenditures are broken out in the data: education, roads and highways, public welfare, health and hospitals, natural resources, public safety, and a catchall – other. These results are presented in Panels A through H of Table 8. As before, we start off with a parsimonious specification which only includes state and year fixed effects and then augment the regression with more controls, including state-specific linear time trends and all socio-economic controls in our fifth and final column.

[Table 8 about here.]

Consistent with the impressions from Figure 2, we find that withholding did not have an effect on state general expenditures, either in the aggregate or when broken out by category. These null results are confirmed if we consider expenses – either as a whole or broken out by category – as a share of personal income, as demonstrated by the null results in Online Appendix Table A.3.

As suggested by Figure 2, a possible explanation for why taxes went up without a similar increase in expenses is the leveling off of state debt. Digging deeper into this initial result, we distinguish between states based on whether they have a strict balanced budget requirement (BBR) or not. Upon splitting the sample into states classified by the Advisory Commission on Intergov-

ernmental Relations (ACIR) as having a strict BBR or not, we observe in Figure 6 that the debt trajectory in states *without* a strict BBR is dramatically impacted by the passage of withholding, whereas no such effects are observed in states without such a requirement.

[Figure 6 about here.]

Regression results consistent with the visual evidence are presented in Table 9 in which we examine whether there was a change in the trajectory of state debt upon the introduction of withholding. The four panels vary the length of the sample window – starting with 6 years on either side of the introduction of withholding (in Panel A) and increasing that to 9 (in Panel B), 12 (in Panel C), and to 15 years (in Panel D).⁵⁰ Regressions reported in columns (1) and (2) are estimated on the sample of states that have a strict BBR whereas regressions in columns (3) and (4) are estimated on the sample of states that do not have such a BBR in place.

[Table 9 about here.]

Comparing the coefficients in columns (1) and (2) with those in columns (3) and (4), we find that the introduction of withholding had a negative and statistically significant impact on the trajectory of state debt but only in the 13 states that lacked a strict BBR. Moreover the degree to which the trajectory of state debt slows down decreases as we expand the length of our window, with the coefficients becoming progressively smaller as we move down the table. In contrast, for the 16 states that had a strict BBR in place, there is no conclusive evidence that state debt levels were impacted by the adoption of withholding.⁵¹ Our interpretation of these results, particularly in light of Figure 2, is that in the pre-withholding period the demand for public spending exceeded the revenue raised by states and debt burdens grew as a result, especially in states without a strict BBR.⁵² However, once states implemented withholding, their revenues and expenses were more in balance and there was a change in the trajectory of state debt. These results also help explain why we did not detect an increase in any category of expenditures; the positive shocks to demand for public spending were building up over several years prior to adoption and that initially manifested itself in the form of an increase in debt.

⁵⁰While increasing the duration of the window makes our estimates of growth rates potentially more reliable, estimates over a longer window may be contaminated with other changes to policy unrelated to the adoption of withholding.

⁵¹While Table 9 considers the ratio of debt to expenses, our results are similar, or stronger, when we operationalize debt as the ratio of debt to revenues or consider the real per capita debt level, i.e. debt in absolute terms.

⁵²Although the presence or absence of a strict BBR appears to matter for debt levels, that fiscal institution does not affect our estimates of the effects of introducing withholding, as shown by RC9 in Table 6.

7 The contribution of withholding to the growth of government

As the final topic of the paper, we would like to revisit Friedman's (1998) claim that withholding provided a major stimulus for the growth in the size of government. We take three approaches to addressing this question. First, we quantify the degree to which withholding may have directly contributed to an increase in personal income taxes and compare that with the overall increase in income taxes during our sample period. Second, we examine if rate increases were more frequent in the post-withholding period relative to the pre-withholding period and whether the sensitivity of rate increases to the electoral calendar changed post-withholding. Third and finally, we compare the growth in the size of government for states that implemented withholding after the adoption of the personal income tax – the states that have formed the basis of our analysis thus far – with a group of states that adopted the personal income tax *and* withholding simultaneously after 1944, states such as Indiana, Michigan, and Pennsylvania. The results from each approach suggest that withholding was *not* the primary driver for the growth in the size of government during the second half of the 20th century; instead it was the demand for revenues that played a leading role in the expansion of government.

7.1 Magnitude of the rise in state personal income taxes on account of withholding

Table 3 shows that personal income taxes increased by 28.4 percent following the adoption of withholding. Although some of this increase likely stemmed from the demand shock, as a first pass, we attribute the entire increase of 28.4 percent to the causal impact of introducing withholding. Thus, because the ratio of income tax revenues to personal income was, on average, 0.82 percent one year before adoption, the response to withholding added 0.23 percentage points.⁵³ During the sample period (1944–2000), the ratio of income taxes to personal income increased from 0.50 percent to 2.67 percent: an increase of 2.16 percentage points. Therefore withholding accounts for only about 11 percent of the overall growth of income tax revenues.⁵⁴ Moreover, the adoption of withholding was most likely the single biggest improvement in the tax collection technology during the period we are studying. Therefore, in terms of empirical relevance, growth in the demand for spending seems to have been a more important factor behind the growth of income taxes than

⁵³Since $28.4 * 0.82 = 0.233$.

⁵⁴Since $0.233/2.16 = 0.108$.

improvements in the ability to collect taxes.⁵⁵

These conclusions are also echoed in Figure A.3 which examines the trend in personal income tax revenues for eight states that adopted withholding during our sample period. To provide some assurance that these observations apply to states from around the country, the states are chosen such that each of the nation's four regions (Midwest, Northeast, South, and West) is represented. Additionally the two Western states are chosen such that they span much of our sample period with Oregon (1948) being the first and California (1971) the second last state to introduce withholding. As we examine the trends for these eight states, we observe that while there was an increase in income tax revenues for each state following the adoption of withholding, that increase pales in comparison to the overall increase in income taxes over our sample period.⁵⁶ Moreover, withholding seems to have resulted in a one-time increase in income tax revenues, rather than resulting in a change in the growth rate of tax revenues itself, something that we formally examine, and find support for, in Table 10.

In Table 10 we examine whether the growth rate of personal income taxes changes once withholding is adopted. The regressions follow the same basic structure as the regressions in Table 3, except that the dependent variable is the change in the logarithm of real per capita personal income tax revenues from the previous year. Given that there was a discrete jump in tax revenues in the year that withholding was introduced, and a transitional dip thereafter (see Figure 3 and the discussion in Section 6.2), to be conservative, we exclude the adoption year and the first two years after adoption. Panels A, B, C, and D vary the length of the sample window – from 6 to 9 to 12 up to 15 years on either side of the introduction of withholding.

[Table 10 about here.]

The results from Table 10 are inconclusive. While all 16 estimates of the impact of withholding on growth rate of the personal income tax that we obtain are positive, 14 are statistically indistinguishable from zero and only two are statistically significant, that too at the 10-percent level. The two statistically significant estimates are obtained when using a 9-year window around

⁵⁵An additional piece of evidence consistent with this view is that top personal income tax rates went up considerably from an average of 5.43 percent in 1944 to 7.01 percent in 2000. Bottom tax rates were significantly lower than top tax rates but they too went up, roughly by a similar magnitude, from 1.31 percent in 1944 to 2.67 percent in 2000 pointing to a growing need for revenues.

⁵⁶That pattern holds up even if we truncate the sample in 1986, omitting the period post passage of the Tax Reform Act of 1986 and the entire decade of the 1990s when robust economic growth and large stock market returns led to a sharp rise in state personal income tax revenues.

adoption (Panel B) and the coefficients become smaller in economic terms and statistical significance as we expand the number of years around the adoption of withholding in Panels C and D. These findings suggest that the effects of withholding on growth rates were weak at best and that they tended to decline over time. Perhaps the most we can infer from the results in Table 10 is that we cannot reject the null hypothesis that the adoption of withholding did not appreciably change the growth rate of personal income taxes. In conjunction with the results from our baseline specification presented in Table 3 (and the robustness checks in Table 6), we conclude that while withholding clearly resulted in a *large, once-and-for-all increase* in personal income tax revenues, it *barely contributed to a change in their long-run trajectory*. The demand for additional revenue likely prompted the adoption of withholding but it also led to changes in policy (such as increases in tax rates or decreases in the income tax exemption threshold (Jensen, 2022)) that drove personal income tax revenues higher both before and after the adoption of withholding. It is the issue of frequency of rate increases we now turn to in our next subsection.

7.2 Frequency of changes to tax rates and base following the adoption of withholding

A popular argument (including one advanced by Milton Friedman) is that the adoption of withholding reduced the visibility of the federal personal income tax and made it easier to enact increases in tax rates. In our sample though, we find that increases in state income tax rates were no more (or no less) likely to occur in the post-withholding period relative to the pre-withholding period. For instance, the top personal income tax rate was increased in 8.9 percent of state-years prior to the introduction of withholding, whereas in the years following the introduction of withholding, tax rate increases occurred in only 6.8 percent of state-years.⁵⁷ Likewise, the bottom personal income tax rate was increased in 7.0 percent of state-years prior to the introduction of withholding, whereas in the years following its introduction, tax rate increases occurred in 6.7 percent of state-years, although neither difference between the proportions is statistically significant.

We take this analysis a step further by examining if the responsiveness of tax policy to the electoral calendar changes once withholding is adopted. The reason we are interested in explor-

⁵⁷Homing in on the period between 1944–1986 does not change this picture either. During the 1944–1986 period, the top income tax rate was increased in 8.4 percent of state-years post-withholding, a *smaller* proportion than in the pre-withholding period.

ing the issue is because the arguments offered by Milton Friedman (among others) suggest that the lower salience of income taxes post-withholding may have enabled politicians to incur lower political costs in increasing taxes once withholding is adopted. Given that increases in tax rates are likely to be unpopular with voters, politicians are less likely to enact such increases during election years.⁵⁸ The question we examine is whether politicians were more open to enacting tax rate increases during election years in the post-withholding period relative to their propensity in the pre-withholding period. We start off by estimating:

$$y_{st} = \gamma_t + \beta_1 \mathbf{1}(ElecYear)_{st} + \lambda_s + \mu_t + \varepsilon_{st}$$

in column (1) of Panel A of Table 11 where y_{st} is an indicator variable set to 1 if the top tax rate goes up and $\mathbf{1}(ElecYear)_{st}$ is an indicator variable for whether it is an election year or not.⁵⁹ Following Finkelstein (2009), we then add in terms corresponding to the presence (or absence) of tax withholding and the interaction thereof with the election year indicator variable.

$$y_{st} = \gamma_t + \beta_1 \mathbf{1}(ElecYear)_{st} + \beta_2 Withholding_{st} + \beta_3 \mathbf{1}(ElecYear)_{st} * Withholding_{st} + \lambda_s + \mu_t + \varepsilon_{st}$$

We re-estimate these specifications adding state-specific linear time trends in columns (3) and (4). Panel B presents results from a similar exercise but here we examine changes to the bottom personal income tax rate.

[Table 11 about here.]

As the negative coefficient on the election year indicator variable in the first row of Panel A conveys, increases in tax rates are less likely to occur during an election year. The introduction of withholding *does not* bring about a fundamental change to the underlying political cycles of tax rate setting: rate changes are no more (or less) likely to occur once withholding is introduced as evidenced by the coefficients in the second row. Most importantly for our purposes though, it is the *responsiveness* of tax rate increases to the electoral calendar that does not change upon the introduction of withholding given that the interaction variable in the third row is statistically insignificant and is, in fact, close to zero. The results in Panel B, examining changes to the bottom

⁵⁸Chang et al. (2021) provide such evidence of politicians shying away from tax increases around elections in the context of the excise tax on gasoline.

⁵⁹A preliminary analysis suggests that having either a gubernatorial or a legislative election reduces the likelihood of an increase in tax rates. Hence, rather than distinguish between gubernatorial and legislative elections, we set an indicator variable equal to 1 when *any* election, either gubernatorial or legislative, takes place that year.

personal tax rate, reaffirm the findings from Panel A. The fact that the *responsiveness* of tax rate increases to the electoral calendar does not change in the post-withholding period is a robust result that holds up to alternative specifications and other empirical approaches, for example, the introduction of two separate indicators for gubernatorial and legislative elections and their interactions, or to using a probit rather than a linear probability model. We obtain similar results when we examine decreases to the personal exemption or decreases to the standard deduction. Such changes, while less frequent than increases in top tax rates, are also less likely to occur in an election year but the introduction of withholding does not either make such changes more frequent or reduce their sensitivity to the electoral calendar.⁶⁰ Our results in this subsection thus lead us to be skeptical of the claim that the introduction of withholding reduced the salience of the state personal income tax and the costs incurred by politicians in enacting increases in tax rates.

Reflecting on our findings in this sub-section and the prior one, they appear to contradict Milton Friedman's claim that withholding provided a major stimulus for the *growth* of income taxes. We add a caveat though that our findings on state income taxes may not apply to the experience with the federal income tax for a number of reasons. First, the responsibilities of the federal and state governments differ markedly; second, they are subject to different institutional constraints (such as balanced budget requirements) and macroeconomic forces; and third, federal income taxes raise considerably more revenue than state income taxes – both as a share of GDP and as a proportion of the overall tax mix.

7.3 Examining the growth in the size of government for states which adopted the personal income tax and withholding simultaneously

Throughout this entire paper, our focus has exclusively been on states that had a personal income tax at the end of World War II and then adopted withholding subsequently between 1948 (Oregon) and 1987 (North Dakota) – states that we characterize as Group 1 states for this subsection. We have shown that although the adoption of withholding led to a large increase in compliance with the personal income tax, it is unlikely to have been a driving factor in leading to an increase in the size of government. Our evidence suggests that state government finances were stressed in the years leading up to the introduction of withholding and although its adoption provided some fiscal space for states, the adoption of withholding did not fundamentally alter the trajectory of

⁶⁰Those additional results are available from the authors on request.

state expenditures or even of state personal income tax revenues. We now take a final attempt at establishing the robustness of this finding by examining the growth in the size of government for a second group of 11 states that adopted withholding *simultaneously* with the introduction of the personal income tax – states such as Indiana, Michigan, and Pennsylvania that we characterize as Group 2 states for this subsection.

To motivate this discussion, we first examine the growth of expenses in the 8 years *following* the adoption of withholding for Group 1 states and compare that with the growth of expenses for Group 2 states in the 8 years *prior* to the adoption of the personal income tax (and withholding) in those states. Upon doing so, we find that while real per capita general expenses grew by 35 percent in Group 1 states from \$712 to \$962 in the 8 years *after* the adoption of withholding, those expenses grew by 54 percent from \$602 to \$924 in Group 2 states in the 8 years *prior* to the adoption of withholding. Thus, the size of government – as measured by real per capita general expenses – grew *faster* in Group 2 states *prior* to the adoption of withholding than in Group 1 states *after* the adoption of withholding. Those results are not sensitive to the precise window employed; if we exclude 1 year prior to and after the introduction of withholding, we find that real per capita general expenses grow by 34 percent for Group 1 states in the *post-withholding* period, whereas they grow by 56 percent for Group 2 states in the *pre-withholding* period. Those conclusions hold up if we operationalize expenses differently, instead expressing them as a share of personal income. By that measure as well, the size of government in Group 2 states expanded *faster* prior to the implementation of withholding than in Group 1 states after the introduction of withholding, regardless of how we construct our window.

The results of this sub-section can be captured in the form of a figure, Figure 7, which confirms the point that the trajectory of increase in general government expenses for Group 2 states *prior* to the implementation of withholding was *faster* than that for Group 1 states *after* the adoption of withholding. These results therefore lead us to be further skeptical of the claim that the adoption of withholding is what causally led to a larger government and instead highlight the importance of a need for government revenues as driving fiscal policy choices.

[Figure 7 about here.]

8 Conclusions

While theories of the size of government have been unanimous in the prediction that improvements

in the ability of governments to collect taxes lead to a bigger government, the empirical literature has so far been inconclusive in attributing the growth of government primarily to improvements in the ability of governments to collect taxes (via higher state capacity) or to an underlying increase in the demand for government. The staggered adoption of withholding at the state level in the U.S. provides a unique setting for us to investigate whether the causality runs from higher state capacity to a larger government or the other way around. It also offers us an opportunity to document the effect of this important policy reform, encompassing the simultaneous introduction of third-party information reporting and withholding, on overall state public finances – something that has not been done in the prior literature.

Using a difference-in-differences estimator, we show that withholding led to a large, immediate, and permanent increase in personal income tax revenues by about 28 percent. We find that the major response of state governments to the improvement in their ability to collect the personal income tax was a shift in the composition of revenue: the share of the personal income tax in state taxes increased by about 24 percent. This result resonates with the view that the political process favors efficient methods of redistribution (e.g., Becker, 1983; Peltzman, 1980; Wittman, 1989).

We also find that withholding produced only a one-time shift in the level of income taxes, but did not accelerate their growth. Likewise, withholding had little impact on government expenses, and, in fact, expenses grew faster for states without a personal income tax in the period *prior to* the introduction of withholding than they grew in states that had a personal income tax in the period *after* its adoption. Overall, the evidence we have suggests that government grew because of the demand for an increase in its size and not because of the mere introduction of withholding.

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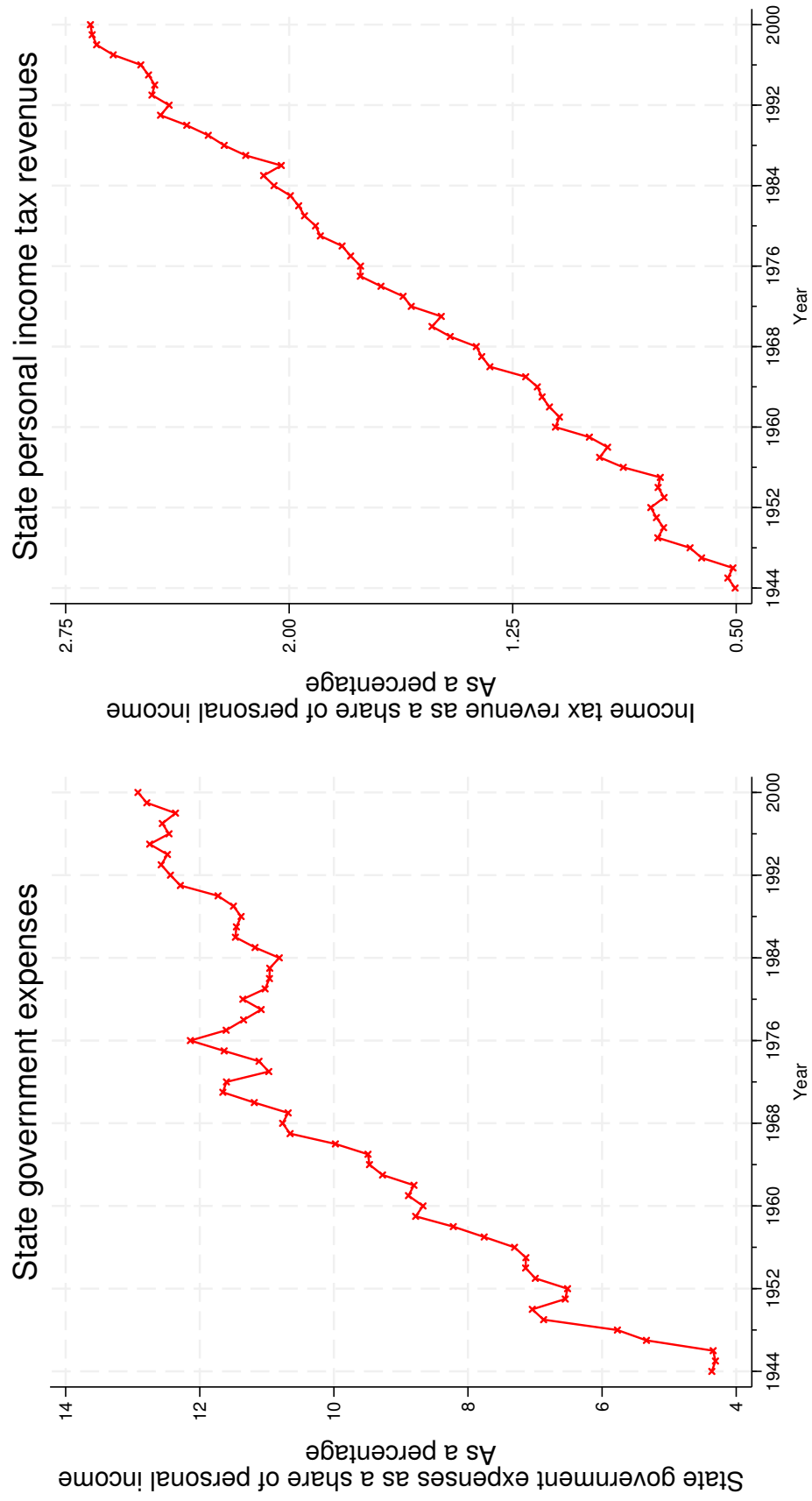
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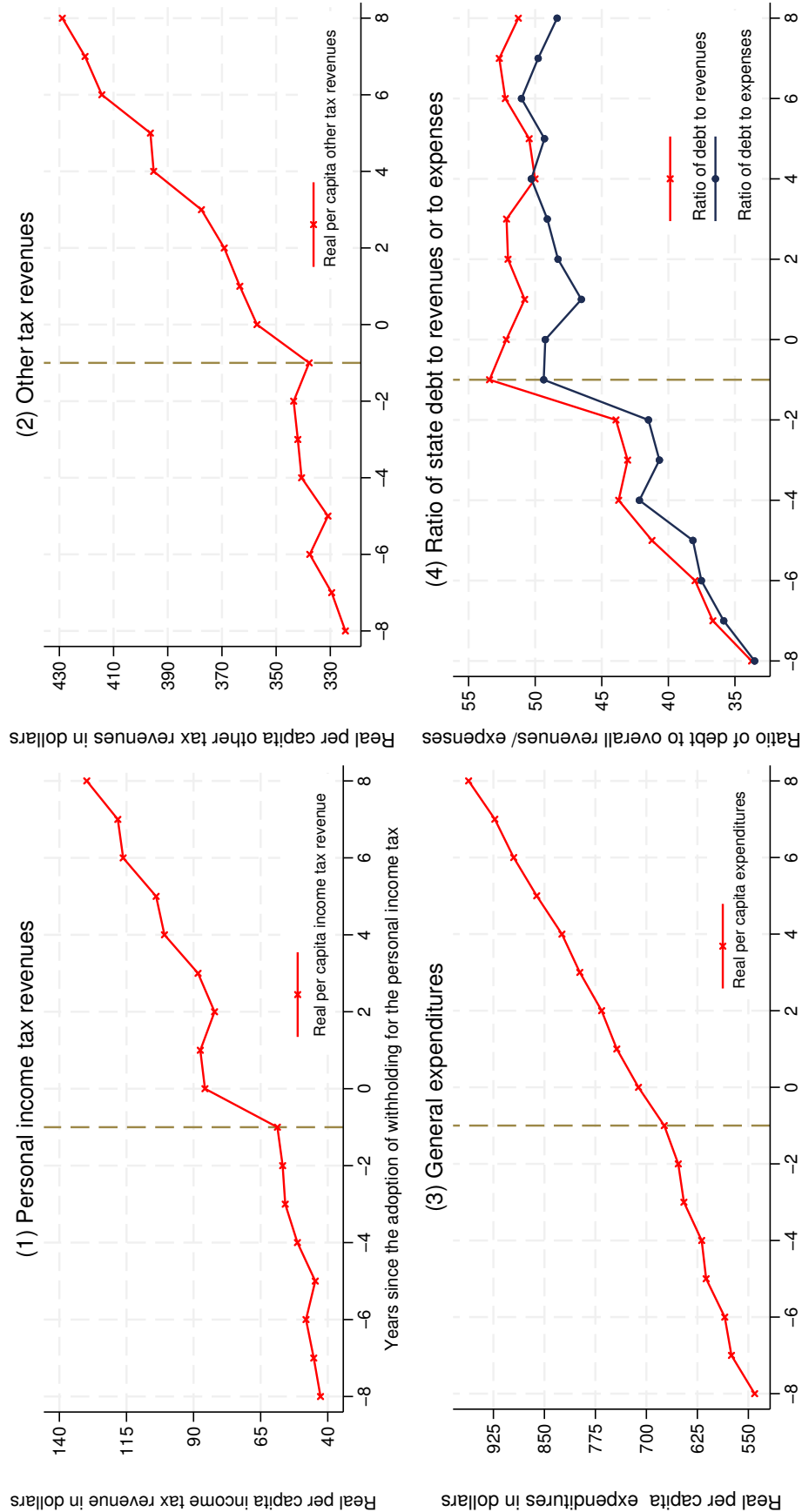
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Figure 1: The size of state government and revenues from the state personal income tax (1944–2000)



Average values for states adopting withholding after introducing the state personal income tax

Figure 2: Trends in state taxes, expenditures, and debt levels around the time of adoption of withholding



Years since the adoption of withholding for the personal income tax

Average value for states which adopted withholding after introducing the state personal income tax

Figure 3: Estimated percentage change in real per capita personal income tax revenues for states adopting withholding after having introduced a state income tax, 95% CI

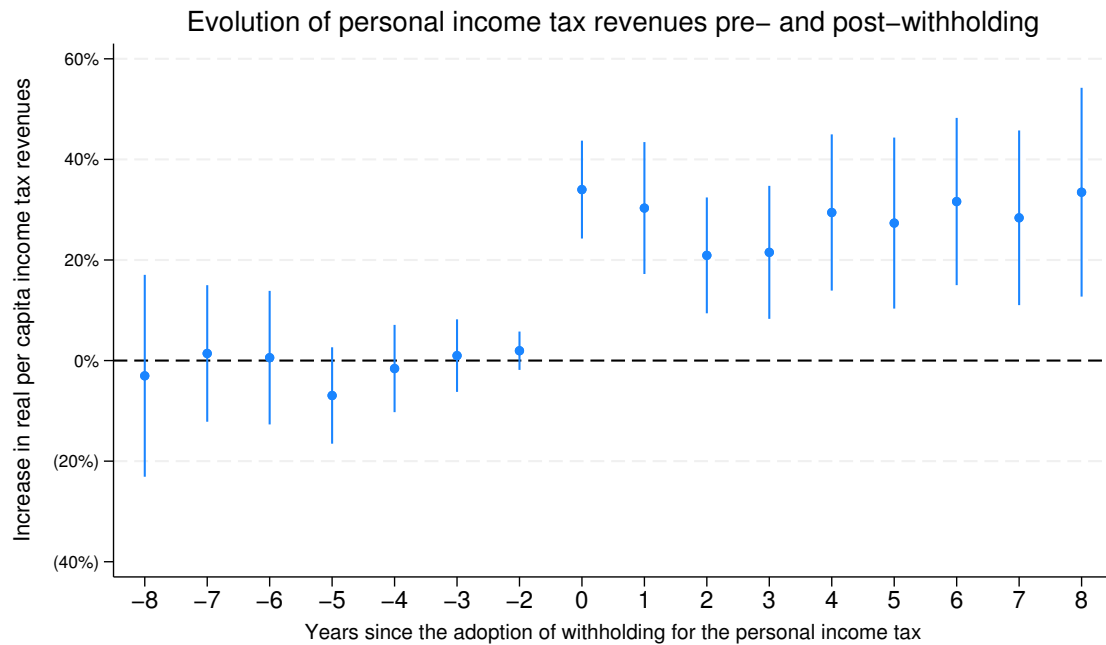


Figure 4: Estimated percentage change in the sum of all tax revenues except for the personal income tax for states adopting withholding after having introduced a state income tax, 95% CI

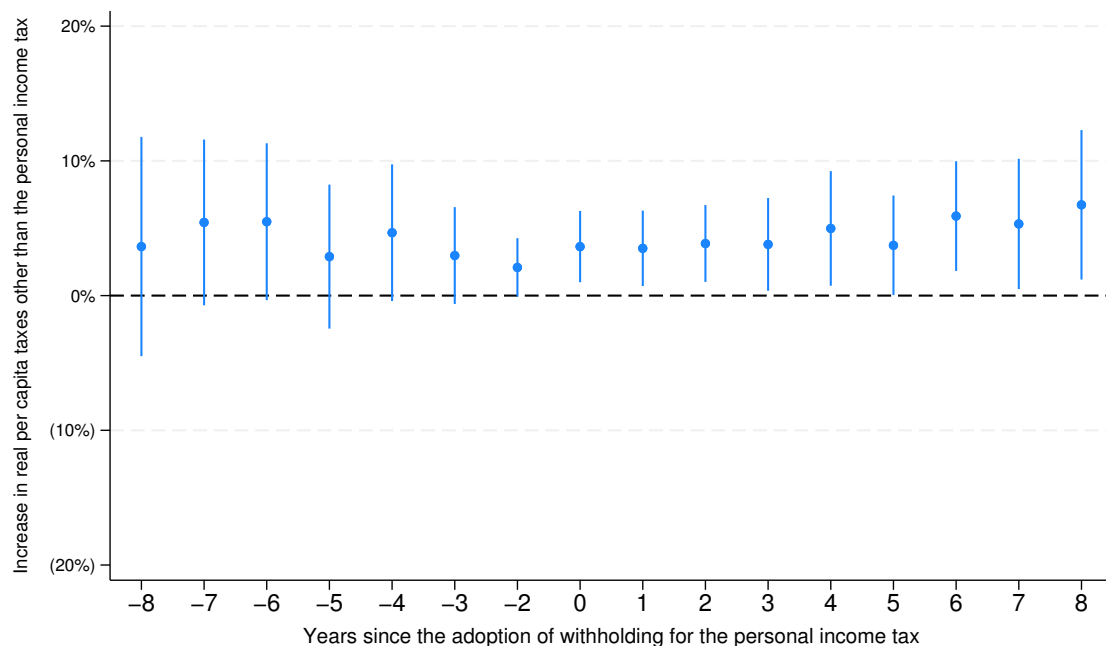
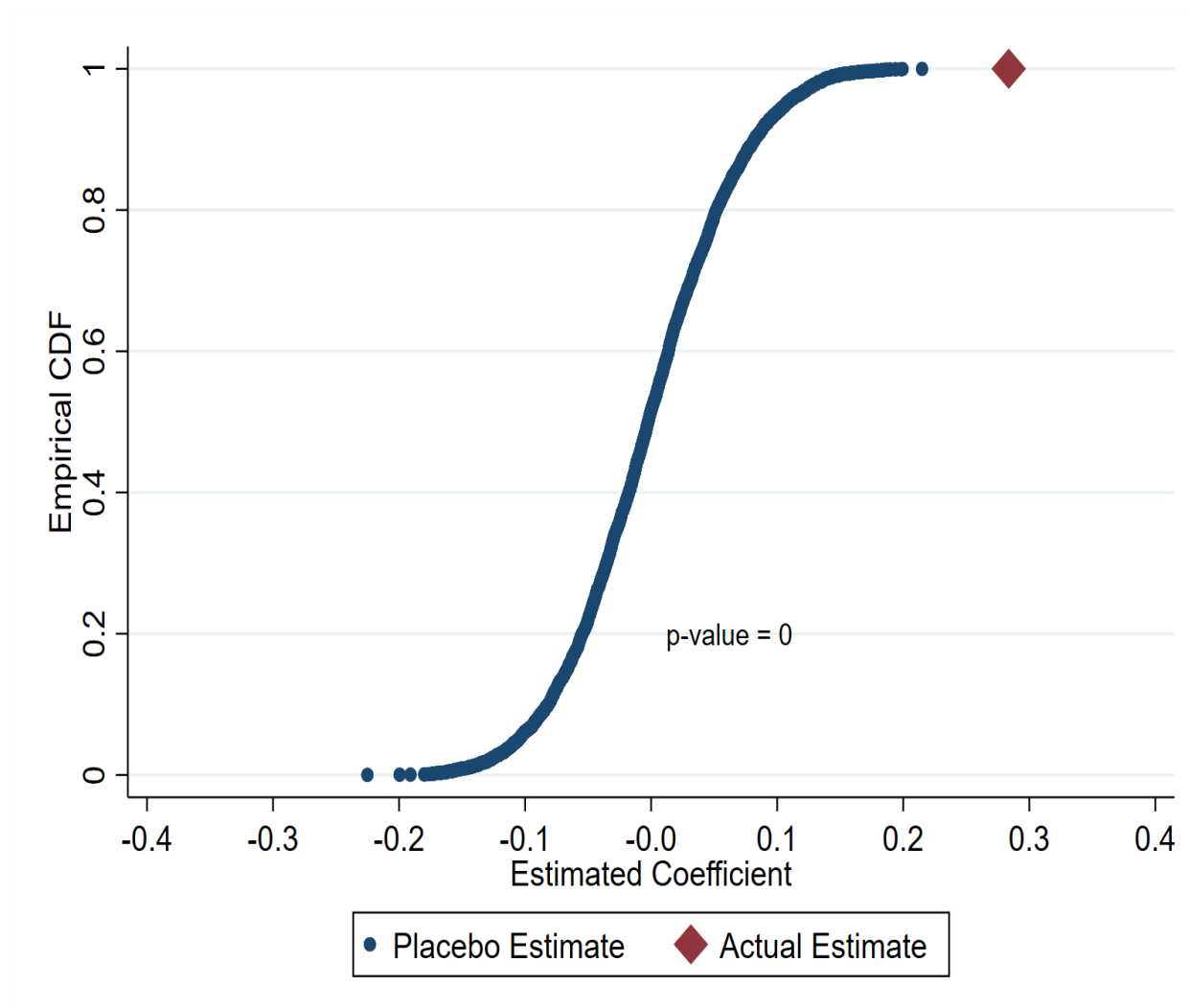


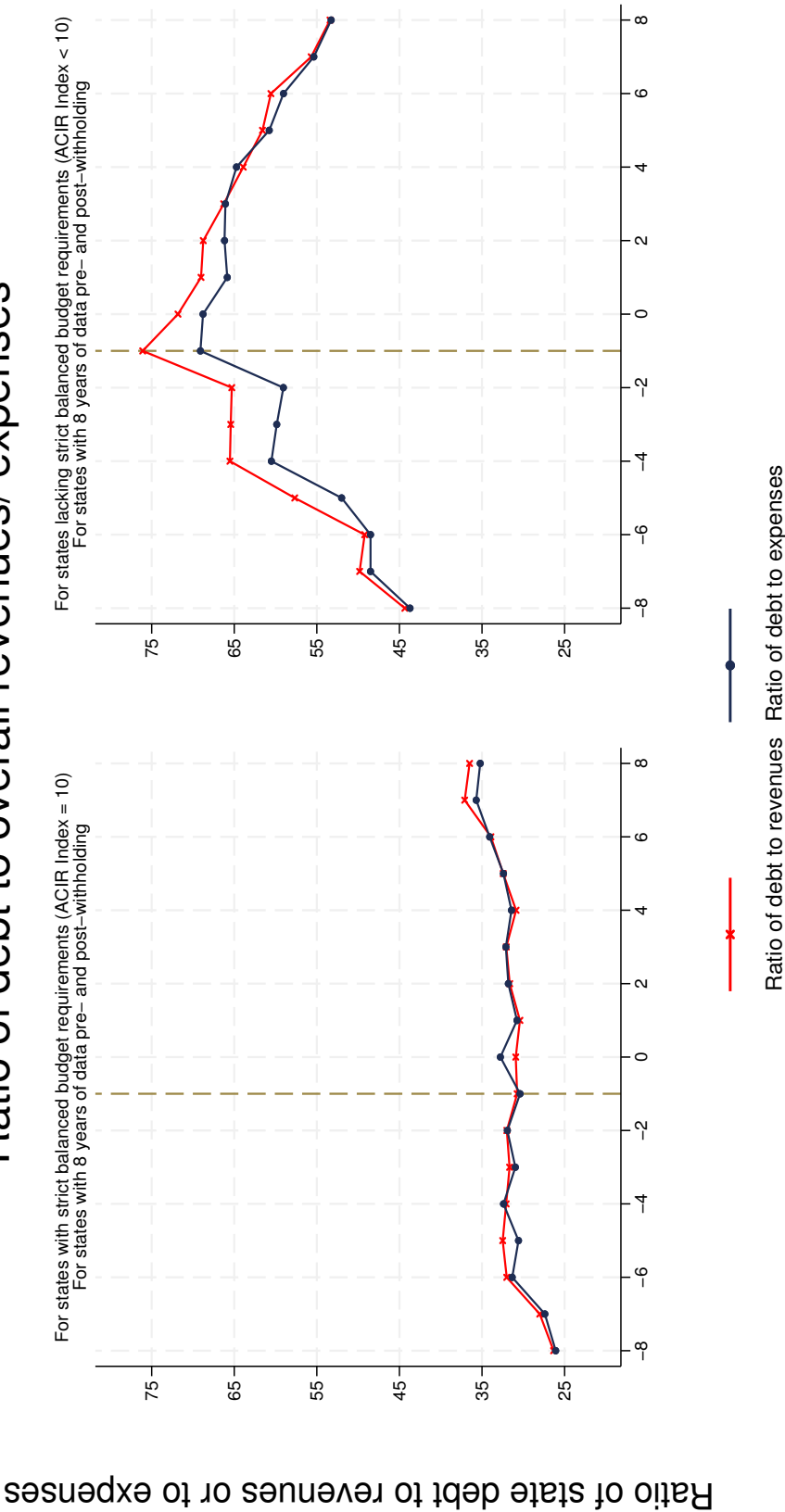
Figure 5: Permutation test: random assignment of the year of adoption



Notes: The placebo adoption years are drawn from a normal distribution, the mean and standard deviation of which are estimated on a support (1948, 1972), i.e. the range of adoption years for all states, except North Dakota (which adopted withholding in 1987). When generating the random placebo years of adoption, the algorithm may pick extreme values indicating the introduction of withholding in say, the year 1920 or 2017. When that happens, we replace such (rare) draws with 1944 (the 1st year in our dataset), and – to keep things symmetric – 1976 (i.e. the last adoption year of 1972 + 4). We then estimate the specification reported in col. (5) of Table 3 and store the estimated coefficient on the withholding dummy. We run 5000 such permutations and plot the cumulative distribution function above, along with the estimated coefficient from Table 3. We report the p-value associated with the null hypothesis that the estimated effect of withholding is drawn from the same distribution that generates the placebo estimates, whose CDF is plotted above. We obtain similar results if we use an uniform distribution for generating the random placebo years of adoption or instead, continue to use a normal distribution but include 1987 (the year of adoption by North Dakota) to compute the mean and standard deviation of the underlying distribution.

Figure 6: Evolution of debt levels for states split based on whether they have balanced budget requirements (BBRs) or not

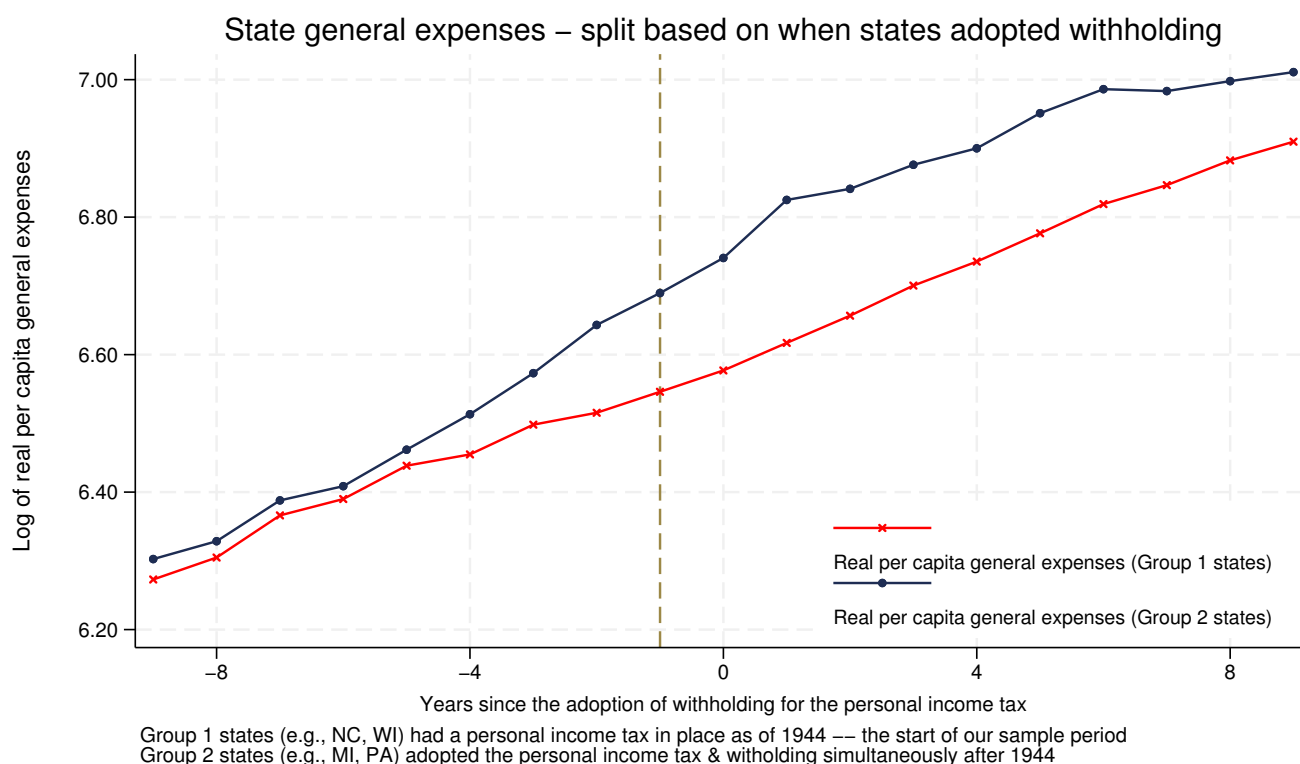
Ratio of debt to overall revenues/ expenses



Years since the adoption of withholding for the personal income tax

The sample of 29 states is split based on whether states have a strict balanced budget requirement (BBR) or not. The data on BBRs are drawn from Bohn and Inman (1996), with the original source being a 1987 report by the Advisory Commission on Intergovernmental Relations (ACIR), "Fiscal Discipline in the Federal System: Experience of the States, Washington, D.C." Of the 29 states in the sample, Bohn and Inman (1996) classify 16 states as having the highest ACIR stringency index of 10 and the remaining 13 states as having a stringency index lower than 10. As the authors note in fn. 10 (p. 23): "The (ACIR) index awards points for whether the rule is: governor must submit (1 point), legislature must pass (2 points), the state may carry a deficit into the next year (4 points), the state cannot carry a deficit into the next biennium budget (6 points), and finally the state cannot carry a deficit into the next fiscal year (8 points). The index of stringency is the points from the "highest" ranked requirement plus one point if the rule is statutory or two point if the rule is constitutional." The value of the ACIR stringency index ranges from 0 (for Vermont – the only state to not have any BBR) to 10 (for 26 states, including, for example, Georgia, and North Carolina). These graphs use a balanced sample, i.e. it only includes states for which we have at least 8 years of data pre- and post-withholding. The reason for imposing this restriction is that the state of Delaware experiences an unusually high rise in its debt from \$7 million in 1948 to \$48 million in 1949 and including Delaware in the analysis results in a plot that is noisier and one we believe is less reliable for drawing inference.

Figure 7: Evolution of real per capita expenses for states split based on when they adopted withholding



Group 1 states includes the 26 states that had a personal income tax in place as of 1944 – the start of our sample period – and for which we have at least 9 years of data prior to and after the introduction of withholding. We exclude Oregon (1948), Delaware (1949), and Vermont (1951) in the construction of this figure (where the adoption years of withholding are provided in brackets) as we lack 9 years of data pre-withholding.

Group 2 states include the 10 states that adopted withholding simultaneously with the introduction of the personal income tax after 1944. These include West Virginia (1961), Indiana (1963), Michigan (1967), Nebraska (1967), Illinois (1969), Maine (1969), Ohio (1971), Pennsylvania (1971), Rhode Island (1971), and New Jersey (1976) where the adoption years of the personal income tax (and withholding) are provided in brackets. We exclude Connecticut which adopted the income tax (and withholding) in 1991 as we lack 9 years of data after the introduction of withholding for the state.

Table 1: Years in which states adopted the personal income tax, income tax withholding, and signed information sharing agreements with the federal government

State	Personal income tax	Withholding	Federal-state information sharing
Oregon	1930	1948	1961
Delaware	1917	1949	1965
Vermont	1931	1951	1965
Arizona	1933	1954	1966
Kentucky	1936	1954	1951
Colorado	1937	1954	1952
Idaho	1931	1955	1964
Montana	1933	1955	1951
Maryland	1937	1955	1963
Alabama	1933	1956	1970
Massachusetts	1916	1959	1963
New York	1919	1959	1963
North Carolina	1921	1959	1950
South Carolina	1922	1959	1964
Utah	1931	1959	1961
Georgia	1929	1960	1968
Oklahoma	1915	1961	1963
Missouri	1917	1961	1962
Minnesota	1933	1961	1957
New Mexico	1933	1961	1963
Louisiana	1934	1961	1971
Wisconsin	1911	1962	1950
Virginia	1916	1963	1963
Arkansas	1929	1966	1963
Kansas	1933	1966	1960
Iowa	1934	1966	1962
Mississippi	1912	1968	1966
California	1935	1971	1961
North Dakota	1919	1987	1964

Source: Tables 1, 18, and 34 of Penniman (1980).

Table 2: Summary statistics (for 29 states with a personal income tax as of 1944; sample period: 1944–2000)

Variable	Obs.	Mean	Standard Deviation	Min.	Max.
Fiscal variables:					
Withholding dummy	1653	0.71	0.45	0	1
Real per capita income tax revenue	1653	177.98	152.33	6.69	805.79
Real per capita corporate income tax revenue	1653	43.38	27.15	0	182.10
Real per capita general sales tax revenue	1653	150.79	111.69	0	494.20
Real per capita selective sales (excise) tax revenue	1653	138.45	36.52	43.29	293.12
Tax Rates:					
Top personal income tax rate	1653	7.14	2.76	1	19.8
Lowest personal income tax rate	1653	1.99	1.17	0.36	6.25
Corporate income tax rate	1653	6.11	2.23	0	12
General sales tax rate	1595	2.80	1.78	0	7
Real motor fuel tax rate (cents per gallon)	1566	9.92	5.51	2	27
Socio-economic characteristics:					
Population (1000s)	1653	3,930	4,678	285	33,988
Real income per capita	1653	9,638	3,582	2,859	21,606
Percent black	1653	11.1	11.8	0.0	47.6
Percent elderly	1653	10.1	2.3	4.5	15.3
Percent young	1653	26.8	4.5	18.9	37.9
Percent urban	1653	62.0	15.7	23.0	94.4
Farm income/ personal income	1653	5.7	6.9	0.0	58.5
Interest & dividend income/ personal income	1653	15.3	4.7	5.0	27.6
Female labor force participation rate	1653	42.8	12.3	18.5	66.0
Share of income going to the top 1 percent of the population	1653	11.0	2.9	5.8	31.9
Average vote share of Democrats in all state-wide races	1653	52.81	10.35	12.86	95.57

Data on the fiscal variables are as described in the text in Section 5. The data on state population, personal income, share of farm income, and the share of interest and dividend income in the state's personal income are from the Bureau of Economic Analysis, Regional Accounts Data. The personal income variables were converted into real 1982 dollars by the consumer price index. The farm income series is very volatile and is negative for some states in some years. Because this variable is meant to capture the long-term agricultural vs. industrial character of the state rather than short-term shocks to agricultural incomes, we use three-year moving averages of farm income in the analysis. The percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas were obtained from Census enumerations. Because Census data are decennial, observation for the years between censuses were generated by simple linear interpolation. Data on variables proxying for demand for revenue, viz., female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races were obtained from Census enumerations, the State-Level Income Inequality Data developed in Frank (2009, 2014), and a dataset originating in the work of Ansolabehere and Snyder (2002) respectively.

Table 3: Estimates of the demand for personal income tax revenue and the mechanical effects of the introduction of withholding on the personal income tax

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Log of real per capita personal income tax revenue						
Withholding dummy	0.392*** (0.0809)	0.293*** (0.0695)	0.328*** (0.0770)	0.335*** (0.0795)	0.284*** (0.0630)	0.217*** (0.0629)
Real income per capita			1.175*** (0.215)	1.302*** (0.229)	0.813** (0.331)	1.155*** (0.274)
Share of farm income			0.138** (0.0595)	0.117** (0.0561)	0.0653 (0.0459)	0.0337 (0.0455)
Share of capital income			0.127 (0.150)	0.180 (0.166)	-0.0954 (0.158)	-0.128 (0.147)
Population			0.0367 (0.100)	0.0563 (0.0945)	-1.006* (0.573)	-1.079** (0.470)
Percent elderly			-0.142 (0.293)	-0.0740 (0.320)	-0.906 (0.547)	-0.436 (0.378)
Percent young			1.137** (0.538)	1.332** (0.543)	2.340** (1.076)	2.126** (0.880)
Percent Black			-0.272*** (0.0650)	-0.290*** (0.0675)	0.00319 (0.108)	0.0858 (0.118)
Percent urban			-0.182 (0.269)	-0.214 (0.282)	-0.403 (0.463)	-0.113 (0.436)
Female labor force participation rate				0.291 (0.392)	0.447 (1.185)	0.600 (1.092)
Income share of the top 1 percent				-0.256 (0.190)	-0.0256 (0.236)	0.102 (0.193)
Average Democratic vote share in all statewide races				0.219 (0.169)	0.296** (0.139)	0.248* (0.122)
Top personal income tax rate						0.451*** (0.0573)
Lowest personal income tax rate						0.0950 (0.103)
State and year fixed effects	Y	Y	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	N	Y	Y
Observations	1653	1653	1653	1653	1653	1653
R-squared	0.93	0.95	0.95	0.95	0.96	0.96

Standard errors clustered by state in parentheses *p < 0.10, ** p < 0.05, *** p < 0.01.

State and year fixed effects are included in all specifications; state-specific linear time trends are included in columns (2), (5), and (6). Columns (3), (4), (5), and (6) include controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas. Columns (4), (5), and (6) include female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding.

Table 4: Estimates of the demand for revenue from taxes other than the personal income tax

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Panel A: Log of real per capita all non-personal income tax revenue					
Withholding dummy	0.0358 (0.0514)	0.00967 (0.0180)	0.0179 (0.0424)	0.0116 (0.0362)	0.0126 (0.0201)
Observations	1653	1653	1653	1653	1653
Dependent variable: Panel B: Log of real per capita corporate income tax revenue					
Withholding dummy	0.150 (0.111)	0.0900 (0.0584)	0.118 (0.102)	0.0719 (0.0872)	0.0836 (0.0668)
Observations	1639	1639	1639	1639	1639
Dependent variable: Panel C: Log of real per capita sales tax revenue					
Withholding dummy	0.103 (0.0698)	0.0257 (0.0339)	0.0806 (0.0578)	0.0707 (0.0571)	0.0184 (0.0289)
Observations	1653	1653	1653	1653	1653
Dependent variable: Panel D: Log of real per capita miscellaneous tax revenue					
Withholding dummy	-0.0559 (0.0655)	-0.00548 (0.0421)	-0.0788 (0.0589)	-0.0681 (0.0612)	0.0139 (0.0424)
Observations	1653	1653	1653	1653	1653
State and year fixed effects	Y	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	N	Y
Socio-economic controls	N	N	Y	Y	Y
Controls capturing demand for tax revenue	N	N	N	Y	Y

Standard errors clustered by state in parentheses *p < 0.10, ** p < 0.05, *** p < 0.01.

State and year fixed effects are included in all specifications; state-specific linear time trends are included in columns (2) and (5). Columns (3), (4), and (5) include controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas. Columns (4) and (5) include female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the same sample as in Table 3, i.e. the 29 states which had a broad-based personal income tax as of 1944, but of those states, not all employ these other sources of revenue or may not have had the corresponding tax in place for the entire period of the sample. For example, although Delaware is included in the sample in Panel B (corporate income tax), the state adopted a corporate income tax in 1957 and it came into effect in 1958.

Table 5: Estimates of the introduction of withholding on the share of taxes collected from the personal income tax

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Log share of taxes collected from the personal income tax					
Withholding dummy	0.339*** (0.0814)	0.250*** (0.0708)	0.295*** (0.0789)	0.296*** (0.0844)	0.238*** (0.0628)
Real income per capita			0.751*** (0.235)	0.914*** (0.239)	0.381 (0.356)
Share of farm income			0.134** (0.0648)	0.111* (0.0638)	0.0629 (0.0448)
Share of capital income			0.262* (0.151)	0.282* (0.155)	-0.172 (0.168)
Population			0.266** (0.121)	0.294*** (0.106)	-0.690 (0.625)
Percent elderly			-0.108 (0.323)	0.0297 (0.320)	-0.536 (0.501)
Percent young			1.126** (0.520)	1.419*** (0.495)	2.391** (1.107)
Percent Black			-0.273*** (0.0787)	-0.269*** (0.0887)	0.0901 (0.122)
Percent urban			-0.156 (0.301)	-0.290 (0.311)	-0.639 (0.455)
Female labor force participation rate				0.540 (0.427)	1.369 (1.264)
Income share of the top 1 percent				-0.291 (0.221)	-0.0879 (0.220)
Average Democratic vote share in all statewide races				-0.00282 (0.180)	0.303** (0.140)
State and year fixed effects	Y	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	N	Y
Observations	1653	1653	1653	1653	1653
R-squared	0.81	0.87	0.85	0.85	0.89

Standard errors clustered by state in parentheses *p < 0.10, ** p < 0.05, *** p < 0.01.

State and year fixed effects are included in all specifications; state-specific linear time trends are included in columns (2) and (5). Columns (3), (4), and (5) include controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas. Columns (4) and (5) include female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding.

Table 6: Robustness checks on estimates of the effect of withholding on personal income tax revenues, non-personal income tax revenues, and share of revenues from the personal income tax

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	Log of real per capita personal income tax revenues		Log of real per capita non-personal income tax revenues		Log share of revenues from the personal income tax	
Baseline Specification	0.293*** (0.0695)	0.284*** (0.0630)	0.00967 (0.0180)	0.0126 (0.0201)	0.250*** (0.0708)	0.238*** (0.0628)
RC1: Pre-trend estimates (falsification exercise: only including pre-adoption years)	-0.0925** (0.0360)	-0.0647* (0.0368)	-0.00234 (0.0192)	0.00753 (0.0143)	-0.0803** (0.0361)	-0.0632 (0.0381)
RC2: Ignoring transitory shocks in years 1 & 2 following adoption	0.293*** (0.0807)	0.271*** (0.0705)	0.0135 (0.0204)	0.0193 (0.0223)	0.246*** (0.0838)	0.221*** (0.0710)
RC3: Including state-specific quadratic time trends	0.299*** (0.0663)	0.293*** (0.0600)	-0.00523 (0.0195)	-0.0131 (0.0196)	0.265*** (0.0667)	0.266*** (0.0627)
RC4: Introducing region-by-year FE	0.305*** (0.0730)	0.288*** (0.0717)	0.00320 (0.0182)	0.0148 (0.0215)	0.267*** (0.0739)	0.243*** (0.0713)
RC5: Introducing division-by-year FE	0.278*** (0.0951)	0.276*** (0.0957)	-0.0120 (0.0192)	-0.00284 (0.0239)	0.259*** (0.0927)	0.253** (0.0931)
RC6.1: Weighted regressions with contemporaneous state population as weights	0.290*** (0.0683)	0.282*** (0.0626)	0.0122 (0.0183)	0.0151 (0.0199)	0.244*** (0.0701)	0.233*** (0.0625)
RC6.2: Weighted regressions with state population in 1944 as weights	0.292*** (0.0689)	0.284*** (0.0634)	0.0117 (0.0184)	0.0150 (0.0201)	0.246*** (0.0708)	0.235*** (0.0626)
RC7: Use of a random effects specification	0.395*** (0.0832)	0.336*** (0.0805)	0.0356 (0.0515)	0.0107 (0.0364)	0.342*** (0.0833)	0.298*** (0.0855)
RC8: Controlling for the presence of a federal-state information sharing agreement	0.295*** (0.0679)	0.286*** (0.0622)	0.00872 (0.0176)	0.0115 (0.0201)	0.253*** (0.0679)	0.241*** (0.0615)
RC9.1: For states with strict balanced budget requirement (ACIR stringency index = 10)	0.295*** (0.0643)	0.321*** (0.0684)	-0.0147 (0.0274)	-0.0218 (0.0273)	0.273*** (0.0666)	0.299*** (0.0651)
RC9.2: For states with no strict balanced budget requirement (ACIR stringency index < 10)	0.329** (0.110)	0.284** (0.102)	0.00345 (0.0269)	0.0213 (0.0319)	0.276** (0.110)	0.219* (0.105)
State and year fixed effects	Y	Y	Y	Y	Y	Y
State-specific linear time trends	Y	Y	Y	Y	Y	Y
Socio-economic controls	N	Y	N	Y	N	Y
Controls capturing demand for tax revenue	N	Y	N	Y	N	Y

Standard errors clustered by state in parentheses * p < 0.10, ** p < 0.05, *** p < 0.01.

Results in this table are variations on the specifications reported in columns (2) and (5) of Tables 3, 4, and 5. The top row of the current table is the baseline specification. Except where noted, all specifications are estimated using an annual, state-level panel of data for the years 1944–2000. A list of the states included in each Census region and division is available at https://www2.census.gov/geo/maps-data/maps/reg_div.txt. Region-by-year FE in RC4 and division-by-year FE in RC5 are for the four Census regions and nine Census divisions respectively. State and year fixed effects and state-specific linear time trends are included in all specifications. Columns (2), (4), and (6) include socio-economic controls for state population that is black, young (less than 15), elderly (65+), and lives in urban areas as well as demand-side factors, viz. female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding.

Table 7: The impact of withholding on tax rates

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Panel A: Log of top personal income tax rate					
Withholding dummy	0.141*	0.152**	0.128**	0.146***	0.134***
	(0.0759)	(0.0674)	(0.0539)	(0.0512)	(0.0479)
Observations	1653	1653	1653	1653	1653
Dependent variable: Panel B: Log of bottom personal income tax rate					
Withholding dummy	0.0185	0.0813	0.0796	0.0809	0.0624
	(0.0975)	(0.0763)	(0.103)	(0.0994)	(0.0513)
Observations	1653	1653	1653	1653	1653
Dependent variable: Panel C: Log of corporate income tax rate					
Withholding dummy	0.178**	0.136***	0.166**	0.153**	0.110**
	(0.0722)	(0.0449)	(0.0673)	(0.0574)	(0.0410)
Observations	1639	1639	1639	1639	1639
Dependent variable: Panel D: Log of general sales tax rate					
Withholding dummy	0.0258	0.00118	0.00455	-0.00704	-0.0124
	(0.0372)	(0.0361)	(0.0326)	(0.0308)	(0.0306)
Observations	1259	1259	1259	1259	1259
Dependent variable: Panel E: Log of excise tax rate on gasoline					
Withholding dummy	-0.00965	0.0150	0.0158	0.0122	-0.000681
	(0.0402)	(0.0354)	(0.0444)	(0.0412)	(0.0394)
Observations	1566	1566	1566	1566	1566
State and year fixed effects	Y	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	N	Y
Socio-economic controls	N	N	Y	Y	Y
Controls capturing demand for tax revenue	N	N	N	Y	Y

Standard errors clustered by state in parentheses *p < 0.10, ** p < 0.05, *** p < 0.01.

State and year fixed effects are included in all specifications; state-specific linear time trends are included in columns (2) and (5). Columns (3), (4), and (5) include controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas. Columns (4) and (5) include female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the same sample as in Table 3, i.e. the 29 states which had a broad-based personal income tax as of 1944, but of those states, not all employ these other sources of revenue or may not have had the corresponding tax in place for the entire period of the sample. For example, although Delaware is included in the sample in Panel C (corporate income tax rate), the state adopted a corporate income tax in 1957 and it came into effect in 1958.

Table 8: Estimates of the effects of withholding on state expenditures

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Panel A: Log of real per capita general expenditures					
Withholding dummy	0.0256 (0.0331)	0.00938 (0.0274)	0.000890 (0.0220)	0.0144 (0.0234)	0.0183 (0.0248)
Dependent variable: Panel B: Log of real per capita spending on education					
Withholding dummy	0.0465 (0.0567)	0.00702 (0.0401)	0.00129 (0.0442)	0.000777 (0.0448)	0.0149 (0.0402)
Dependent variable: Panel C: Log of real per capita spending on roads					
Withholding dummy	-0.00969 (0.0621)	-0.0739 (0.0589)	-0.0569 (0.0527)	-0.0580 (0.0549)	-0.0690 (0.0640)
Dependent variable: Panel D: Log of real per capita spending on welfare					
Withholding dummy	0.0148 (0.107)	-0.0442 (0.132)	-0.0335 (0.105)	0.0316 (0.118)	-0.0820 (0.114)
Dependent variable: Panel E: Log of real per capita spending on health					
Withholding dummy	0.0476 (0.0505)	-0.0369 (0.0273)	-0.0193 (0.0387)	-0.0200 (0.0385)	-0.0538* (0.0298)
Dependent variable: Panel F: Log of real per capita spending on natural resources					
Withholding dummy	-0.0309 (0.0775)	-0.0647 (0.0598)	-0.0653 (0.0654)	-0.0722 (0.0595)	-0.0660 (0.0576)
Dependent variable: Panel G: Log of real per capita spending on public safety					
Withholding dummy	0.00240 (0.0490)	-0.0395 (0.0330)	-0.0634** (0.0305)	-0.0499 (0.0336)	-0.0587** (0.0255)
Dependent variable: Panel H: Log of real per capita spending on other categories					
Withholding dummy	-0.0666 (0.0628)	-0.000132 (0.0591)	-0.0616 (0.0532)	-0.0226 (0.0519)	0.0243 (0.0455)
Observations (for all panels)	1653	1653	1653	1653	1653
State and year fixed effects	Y	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	N	Y
Socio-economic controls	N	N	Y	Y	Y
Controls capturing demand for tax revenue	N	N	N	Y	Y

Standard errors clustered by state in parentheses *p < 0.10, ** p < 0.05, *** p < 0.01.

State and year fixed effects are included in all specifications; state-specific linear time trends are included in columns (2) and (5). Columns (3), (4), and (5) include controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas. Columns (4) and (5) include female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding.

Table 9: Estimates of the growth rate in the ratio of debt to expenditures pre- and post-withholding for windows of varying lengths

	(1)	(2)	(3)	(4)
Dependent variable: Annual change in the log of ratio of debt to general expenses				
Panel A: +/- 6-year window around adoption				
	States with strict BBR		States without strict BBR	
Withholding dummy	0.0151	0.0528	-0.153*	-0.150**
	(0.0791)	(0.0969)	(0.0716)	(0.0649)
Observations	207	207	166	166
Panel B: +/- 9-year window around adoption				
Withholding dummy	-0.0308	-0.0187	-0.117**	-0.125**
	(0.0546)	(0.0758)	(0.0518)	(0.0569)
Observations	300	300	239	239
Panel C: +/- 12-year window around adoption				
Withholding dummy	-0.0501	-0.0389	-0.0826**	-0.0816**
	(0.0634)	(0.0727)	(0.0332)	(0.0339)
Observations	384	384	310	310
Panel D: +/- 15-year window around adoption				
Withholding dummy	-0.0607	-0.0461	-0.100**	-0.0797*
	(0.0542)	(0.0564)	(0.0358)	(0.0387)
Observations	459	459	376	376
State and year fixed effects	Y	Y	Y	Y
Socio-economic controls	Y	Y	Y	Y
Controls capturing demand for tax revenue	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	Y

Standard errors clustered by state in parentheses *p < 0.10, ** p < 0.05, *** p < 0.01. State and year fixed effects are included in all specifications, as are all socioeconomic controls and controls capturing demand for tax revenue. For a complete list of controls, refer to notes to Table 3. Columns (2) and (4) include controls for state-specific time trends. The dependent variable is the annual change in the log of ratio of debt to general expenses. The unit of observation is a state-year. Regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding. The sample is however split based on whether states have a strict balanced budget requirement (BBR) or not, drawing from Bohn and Inman (1996). The original source for the data is a 1987 report by the Advisory Commission on Intergovernmental Relations (ACIR), "Fiscal Discipline in the Federal System: Experience of the States, Washington, D.C." Of the 29 states in our sample, Bohn and Inman (1996) classify 16 states as having the highest ACIR stringency index of 10 and the remaining 13 states as having a stringency index lower than 10. As the authors note in fn. 10 (p. 23): "The (ACIR) index awards points for whether the rule is: governor must submit (1 point), legislature must pass (2 points), the state may carry a deficit into the next year (4 points), the state cannot carry a deficit into the next biennium budget (6 points), and finally the state cannot carry a deficit into the next fiscal year (8 points). The index of stringency is the points from the "highest" ranked requirement plus one point if the rule is statutory or two point if the rule is constitutional." The value of the ACIR stringency index ranges from 0 (for Vermont – the only state to not have any BBR) to 10 (for 26 states, for example, Georgia).

Table 10: Estimates of the growth rate in personal income taxes pre- and post-withholding for windows of varying lengths

	(1)	(2)	(3)	(4)
Dependent variable: Annual change in the log of per capita personal income tax revenues				
Panel A: +/- 6-year window around adoption				
Withholding dummy	0.0166 (0.0409)	0.0358 (0.0557)	0.0179 (0.0418)	0.0597 (0.0519)
Observations	271	271	271	271
Panel B: +/- 9-year window around adoption				
Withholding dummy	0.0534 (0.0315)	0.0412 (0.0305)	0.0492* (0.0288)	0.0473* (0.0272)
Observations	440	440	440	440
Panel C: +/- 12-year window around adoption				
Withholding dummy	0.0262 (0.0276)	0.0382 (0.0285)	0.0236 (0.0273)	0.0408 (0.0280)
Observations	595	595	595	595
Panel D: +/- 15-year window around adoption				
Withholding dummy	0.0249 (0.0263)	0.0260 (0.0269)	0.0214 (0.0258)	0.0278 (0.0231)
Observations	736	736	736	736
Years dropped	Year of adoption and the 1st and 2nd post-adoption years			
State and year fixed effects	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	Y
Socio-economic controls	N	N	Y	Y
Controls capturing demand for tax revenue	N	N	Y	Y

Standard errors clustered by state in parentheses *p < 0.10, ** p < 0.05, *** p < 0.01.

State and year fixed effects are included in all specifications, and state-specific trends are included in columns (2) and (4). Columns (3) and (4) include controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas as well as three controls capturing demand for tax revenue, viz. female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. The dependent variable is the annual change in the log of real per capita state personal income tax revenues. Fiscal and socio-economic variables are in logs. The unit of observation is a state-year. Regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding.

Table 11: The impact of withholding on the existence of political cycles of tax rate increases

	(1)	(2)	(3)	(4)
Panel A: Dependent variable: Indicator variable equals 1 if there is an increase in the top personal income tax rate; 0 otherwise				
Any election year	-0.0409*** (0.0130)	-0.0684** (0.0277)	-0.0396*** (0.0131)	-0.0665** (0.0273)
Dummy for withholding		-0.0397 (0.0504)		-0.0599 (0.0548)
Any election year*		0.0397 (0.0345)		0.0394 (0.0343)
Dummy for withholding				
Observations	1653	1653	1653	1653
"R-squared"	0.098	0.100	0.12	0.12
Panel B: Dependent variable: Indicator variable equals 1 if there is an increase in the bottom personal income tax rate; 0 otherwise				
Any election year	-0.0240*** (0.00860)	-0.0374* (0.0204)	-0.0229** (0.00838)	-0.0353* (0.0204)
Dummy for withholding		-0.0138 (0.0415)		-0.0164 (0.0453)
Any election year*		0.0192 (0.0258)		0.0179 (0.0261)
Dummy for withholding				
Observations	1653	1653	1653	1653
"R-squared"	0.077	0.078	0.095	0.095
State and year fixed effects	Y	Y	Y	Y
Socio-economic controls	Y	Y	Y	Y
Controls capturing demand for tax revenue	Y	Y	Y	Y
State-specific linear time trends	N	N	Y	Y

Standard errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%

State and year fixed effects are included in all specifications; state-specific linear time trends are included in columns (3) and (4). All columns include controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas, areas as well as demand-side factors, viz. female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding.

Data for whether a gubernatorial election takes place in any given year or not is constructed based on data from the Congressional Quarterly Voting Collection. Data on whether an election takes place to the State Assembly or not is based on "State Partisan Balance Data, 1937 - 2011", hdl:1902.1/20403, Harvard Dataverse, V1 (Klärner, 2013). Our independent variable – any election year – is set to 1 if either a gubernatorial election or a legislative election takes place in that year.

Online Appendix

A Addressing Concerns about Two-way Fixed Effects Estimators

A large and growing literature has pointed out the potential pitfalls of using two-way fixed effects estimators (e.g., Athey and Imbens (2022), Borusyak, Jaravel, and Spiess (2021), Callaway and Sant’Anna (2021), Goodman-Bacon (2021), and Sun and Abraham (2021)). Our reading of this literature gives us reason to pause and reflect on whether the underlying assumptions justifying the use of the two-way fixed effects estimator (TWFE) hold in our context but in the end, we believe that the TWFE estimator does provide us an unbiased estimate of the increase in real per capita income tax revenues following the adoption of withholding. We make that claim for a number of reasons anchoring our discussion with reference to the decomposition in Goodman-Bacon (2021):

$$\text{plim} \hat{\beta}^{DD} = \beta^{DD} = VWATT + VWCT - \Delta ATT \text{ (ref: equation (15), p. 7).}$$

In this expression, $VWATT$ is the “variance-weighted average treatment effect on the treated”, $VWCT$ are “variance-weighted common trends” which generalizes common trends to a setting with timing variation, and ΔATT equals a weighted sum of the change in treatment effects within each timing group’s before and after a later treatment time. For the reasons that we argue below, the last two terms in the above equation can be ignored and the TWFE estimator does recover an average treatment effect on the treated, albeit one that is variance-weighted with weights depending on the “absolute size of the subsample” (number of units undergoing treatment) and “variance of the fixed-effects-adjusted treatment variable in the subsample”. To the extent that all the ATTs are the same, variance-weighting does not matter.

First, as Figure 3 suggests, there was no unobservable trend in personal income tax revenues before the adoption of withholding given that the leads for the pre-withholding period are indistinguishable from zero. Thus, we contend that the common trends assumption are met resulting in the second term of the above expression, $VWCT$, equaling zero.

We also argue that the third term in the decomposition, ΔATT , can be ignored in our context. As Goodman-Bacon (2021) notes “time-varying effects bias estimates away from $VWATT$ because $\Delta ATT \neq 0$.” Here however, we find little evidence of time-varying effects because when we decompose the post-withholding years into three periods, we do not discern any statistically significant differences between the short-, medium-, and long-run estimates. The results from that decomposition exercise indicate that the effects of withholding do *not* grow larger with time. Additionally, our formal examination of the growth rate in per capita income pre- and post-withholding

presented as Table 10 indicates no change in growth rates following the adoption of withholding. Thus, withholding causes a discrete jump *but not* a change in the growth rate of revenues. Lastly, a sample-split between early and late adopters of withholding also indicates no statistically significant differences across the sub-samples of states suggesting that the treatment outcome is *not* influenced by the duration of treatment. Those pieces of evidence, taken as a whole, offer some confidence that the term, ΔATT , stemming from heterogeneity in treatment effects over time, can be ignored in our context.

Although we have evidence to suggest that the two terms, $VWCT$ and ΔATT , can be assumed to be zero in our context, we also subject our results to three robustness checks provided in Goodman-Bacon (2021) which re-examines prior work regarding the impact of unilateral divorce laws. Although Stevenson and Wolfers (2006) find that the passage of unilateral divorce laws leads to a drop in female suicide rates by about 9.7 percent, Goodman-Bacon (2021) finds those results not robust to either the inclusion of state-specific linear time trends or region-by-year fixed effects or weighting states by population (Refer Table 2, p. 15). In contrast, our results in Table 3 include state-specific linear time trends, and, as demonstrated by the results in Table 6, they are robust to introducing region-by-year fixed effects (RC4) and weighting states by population, either using contemporaneous population (RC6.1) or using population as of 1944 as weights (RC6.2). Additionally, they are robust to the inclusion of state-specific quadratic time trends.

Although we believe – and have attempted to justify that belief – that the TWFE estimator is valid in our context, we also confirm our results through the use of an estimator that is robust to heterogeneous or dynamic treatment effects and was developed to address concerns with the TWFE. Using the “did_multiplegt” module (de Chaisemartin and D’Haultfoeuille, 2020), we estimate an immediate impact of about 35 percent in tax collections following the adoption of withholding. Moreover, when we include leads in our empirical specification to test for parallel trends, we confirm our result that the leads are indistinguishable from zero. Lastly, we generate an event study plot using this package and include it as Figure A.4. The figure confirms the lack of a pre-trend prior to the adoption of withholding and a sharp rise right when withholding is adopted, consistent with the message from Figure 3 which was generated using the conventional TWFE.

B Case Studies of Adoption

The following section offers three additional case studies of the circumstances in which withholding was introduced in the states of Oregon, Vermont, and Minnesota. Along with the example of

California provided in the main text, these states span the gamut in terms of when they adopted withholding, with Oregon being the first, and California the second last state to introduce withholding. Additionally, these states are drawn from different regions, with Oregon and California belonging to the West, Vermont to the Northeast, and Minnesota to the Midwest. In spite of these differences, what the case studies illustrate is that the passage of legislation enabling the introduction of withholding was rarely uncontroversial and often hinged on the personalities involved, rather than the underlying merits (or demerits) of the policy or the specific fiscal needs of the state. In particular, the identity and views of the Governor, along with those of the Chair and the Ranking Member of the state House and Senate Finance Committees, along with their willingness to compromise, ended up playing an outsized role in each state's decision as to when to adopt withholding. These case studies illustrate that the timing of the introduction of withholding was a function of idiosyncratic political circumstances, reminiscent of the findings of Penniman (1980, Ch. 1) who makes a similar observation in the context of introductions of the personal income tax by U.S. states.

Oregon's experience with the introduction of withholding: A decision by the State Supreme Court saved withholding In 1948 Oregon became the first state to introduce withholding of the personal income tax for all residents. Although the early passage of the law might give the impression that adoption of withholding in the state was uncontroversial, that turns out to be untrue. In April 1947 the Oregon state legislature approved a measure that would impose income tax withholding (and a reduction in personal exemptions) effective January 1, 1948, should a referendum on a state sales tax, to be levied at 3 percent, and scheduled for October 1947 fail to pass.⁶¹ When Oregon voters rejected the referendum imposing a state sales tax on October 7, 1947, thereby automatically putting withholding into effect from January 1, 1948, the Oregon Federation of Labor (OFL) initiated a lawsuit against the state tax commission. The OFL asked that the two acts be declared unconstitutional on the grounds that their implementation had been made contingent upon the outcome of the sales tax vote, and also on the grounds that the legislature had attempted to prevent a referendum against the income tax measures. Although a Marion county circuit court judge ruled against the plaintiffs on November 24, 1947, the plaintiffs appealed this decision to the state Supreme Court.⁶² Ultimately on December 18, 1947, less than two weeks before the withholding provisions were supposed to come into effect, the Oregon Supreme Court ruled that

⁶¹"Nearly 600 Laws Enacted by Longest Legislature," p. 10, Statesman Journal, 06 April 1947

⁶²"Court Upholds New Income and Withholding Tax Acts," p. 1, Albany Democrat-Herald, 24 Nov. 1947.

the legislative acts introducing withholding of the income tax (and a reduction in exemptions) had been constitutional⁶³ and nothing in the state Constitution prevented the legislature from making these provisions contingent on the outcome of the sales tax vote as had been alleged by plaintiffs.

Vermont's experience with the introduction of withholding: A few false starts first Although Vermont was the third state to have adopted withholding, its introduction in 1951 too followed controversy and a number of prior failures. An initial attempt to introduce withholding in 1947, initiated by the newly-elected Governor Ernest W. Gibson Jr.,⁶⁴ cleared the state House but failed in the state Senate. A resulting compromise between the two chambers however resulted in the adoption of withholding for *non-residents*.⁶⁵ A subsequent attempt in 1949 to introduce withholding more broadly again had the support of the Governor and cleared the state House but opposition to the measure in the state Senate doomed its passage, marking the second time that the legislature had refused to introduce withholding for all residents.⁶⁶ The ultimate passage of withholding legislation two years later was more uncertain because unlike Governor Gibson, his successor, the newly-elected Lee Emerson, had been less supportive of withholding.⁶⁷ Nonetheless the next attempt at introducing withholding in 1951 did succeed and with that Vermont became only the third state in the country to have a broad-based withholding scheme in place.

Minnesota's experience with the introduction of withholding: Numerous false starts here as well As was the case with Vermont, the introduction of withholding was not a straightforward endeavor in Minnesota – a state which introduced withholding in 1961, close to the median year of adoption in our sample. As early as 1955, a proposal to introduce withholding had passed the House, with the strong backing of Governor Orville Freeman, only to fail in the state Senate.⁶⁸ In response to this failed attempt to introduce withholding, the Governor set up an interim legislative commission to study the issue but when the 15-member commission issued its report in 1956, the commission narrowly voted 8-7 *against* the introduction of withholding in Minnesota.⁶⁹ Notwithstanding the commission's majority report advocating against the adoption of withholding, Governor Freeman again proposed introducing withholding, first in 1957⁷⁰ and then in 1959,⁷¹ only

⁶³"State Income Tax Boost, Wage Deductions Upheld," p. 1, Albany Democrat-Herald, 19 Dec. 1947.

⁶⁴"Steeper State Income Tax is Given Support," p. 1, White River Valley Herald, 27 Mar 1947

⁶⁵"Income Tax Withholding Explained," p. 1, The Manchester Journal, 18 Dec 1947

⁶⁶"Income Tax Bill Voted Yesterday," p. 1, Montpelier Evening Argus, May 14 1949

⁶⁷"For Withholding Tax," p. 4, Springfield Reporter, 11 Apr 1951

⁶⁸"Legislature Enters 3rd Day of Deadlock," p. 1, The Minneapolis Star, 23 Apr 1955

⁶⁹"Withholding Falts," p. 18, The Minneapolis Star, 08 Aug 1956

⁷⁰"Withholding Plan Spiked in Senate," p. 1, The Winona Daily News, 12 Apr 1957

⁷¹"Senate Tax Group Rejects Withholding," p. 1, The Winona Daily News, 09 May 1959

to have those measures clear the state House but fail in the state Senate. Ultimately the adoption of a plan to forgive the prior year's (1960) tax liability by 75 percent, similar to what the federal government adopted when withholding was first introduced, softened the opposition and led to its adoption by the General Assembly and its approval by newly-elected Governor, Elmer L. Andersen, in 1961: *six* years after it had been first introduced in the state legislature.⁷²

C Additional robustness checks

In this Section, we check robustness of the finding that personal income tax revenues and the share of revenues from the personal income tax increased following the adoption of withholding using a number of tests.⁷³ To varying degrees they address the possibility that there are omitted variables in our analysis which induce an upward bias on the estimated effect of withholding. In the interest of space, we only present the coefficient on this dummy variable in Table 6, with complete results including coefficients on the control variables available on request.

RC1: Pre-trend estimates Beyond the two approaches outlined in Section 6.2 to address the issue of pre-existing trends, we illustrate a third approach here. We examine the trend in revenues from the personal income tax prior to the introduction of withholding by discarding all observations after the adoption of withholding and introducing a dummy variable for observations either one or two years before adoption, with the other pre-withholding years being absorbed into the constant. The results, presented as Robustness Check 1, reveal no significant positive pre-trend; in fact, the coefficients on the dummy variable corresponding to observations one or two years before the adoption of withholding are negative. These results reduce the likelihood that the baseline estimates we present in Table 3 are driven by preexisting trends in revenue; personal income tax revenues were possibly declining prior to the introduction of withholding.

RC2: Ignoring transitory shocks in the first two years after adoption As described in Section 6.2, states experienced a windfall following the adoption of withholding because of a one-

⁷²“Withholding Top Result of Session,” p. 1, The Winona Daily News, 09 Jun 1961

⁷³ For completeness, we include identical robustness checks for revenues from all other taxes as well.

time forward shift in the remittance of taxes. This effect appears in the first and second year post adoption because in many states withholding came into effect on January 1, while in all but two states of our sample (Alabama and New York) the fiscal year begins on July 1. Thus the forward shift in tax collections that occurs in the first calendar year after adoption is accounted for in two fiscal years and to reassure the reader of our results, we present this robustness check in which we drop the adoption year and the first two years after the introduction of withholding.

RC3: Including state-specific quadratic time trends To account for the strong growth in state tax revenues over the sample period, as well as changes in the structure of the economy such as the share of employees in the labor force and favorable economic trends, we include state-specific linear time trends thus allowing tax revenues to evolve separately in each state where withholding was introduced. However, to the degree that the underlying growth in state taxes is convex, the inclusion of state-specific linear trends could lead to overestimating the ‘post-withholding’ dummy. To address this concern, we implement state-specific quadratic time trends but, in fact, we find even larger estimates compared to those with linear time trends. In any event, this check confirms that the introduction of withholding resulted in a large and substantial increase in personal income tax revenues.

RC4: Including region-by-year fixed effects States that adopted withholding after they already had adopted a personal income tax were spread across all four U.S. Census regions. In a demanding robustness check, we introduce region-by-year fixed effects. Our identification thus comes from comparing means for a given state before and after the adoption of withholding allowing for each region to be hit by a unique shock each year and additionally allowing revenues in each state to evolve separately over time. We find our results for the personal income tax robust to the inclusion of these region-by-year fixed effects.

RC5: Including division-by-year fixed effects In another robustness check that is even more demanding, we introduce division-by-year fixed effects for each of the nine Census divisions and estimate the impact of withholding. Even though some states drop out of the estimation (because there was only one state in that division which had a personal income tax in 1944),⁷⁴ the estimated

⁷⁴For example, in the East Central region that includes the states of Illinois, Indiana, Michigan, Ohio, and Wisconsin, only Wisconsin had a personal income tax at the start of our sample period in 1944. While the remaining states have a personal income tax today, they did not at the start of our sample period. The same comment applies also to the Middle Atlantic region, that includes the states of New York, New Jersey, and Pennsylvania, with the latter two states introducing the personal income tax (and withholding) in 1971 and 1976 respectively.

impact of withholding is similar to those in our baseline.

RC6: Weighted regressions where the state population serves as weights Our regressions thus far have been unweighted under the premise that the adoption of withholding in each state can be viewed as a policy change of equal importance, regardless of the size of those states. In this check we weight observations by state population but find that our results are practically unchanged. To address any potential concerns about in- and out-migration because of a state's fiscal policies, as suggested by Cassidy et al. (2024), as an additional check, we use weights based on the population at the start of our sample period in 1944. Making that change however does not affect our results assuaging concerns that readers may have about mobility responses of taxpayers.

RC7: Use of a random effects specification Our results thus far have included state fixed effects to partial out time-invariant differences that are likely to influence tax compliance. That choice is justifiable on the basis of a Hausman test which rejects the null hypothesis of no correlation between unobserved state-specific random effects and the explanatory variables. However, here we estimate a random effects specification as Griliches and Hausman (1986) stress that observing similar estimates across alternative estimation techniques signals the absence of serious errors-in-variables problems. Using the random effects specification, we find that the adoption of withholding had an even larger impact of about 34 percent in the most complete specification that includes state-specific linear time trends and several time-varying covariates.

RC8: Controlling for the presence of federal-state information sharing agreements In addition to adopting withholding, another policy reform that is likely to have improved compliance with the personal income tax involved state governments signing agreements with the federal government to exchange information about income tax audit plans and techniques. Under these agreements that were signed between the 1950s and the 1970s, states would receive revenue agent reports from the IRS that summarized the conclusions from auditing both individual and corporate tax returns (Penniman, 1980). Recent work (Troiano, 2018) suggests that these agreements increased state income tax revenues. In a robustness check, we control for whether an intergovernmental agreement existed between the state and the federal government and find that although signing an agreement led to an additional increase of about 11 percent in collections, the effects of introducing withholding are robust at about 29 percent, similar to our baseline results.

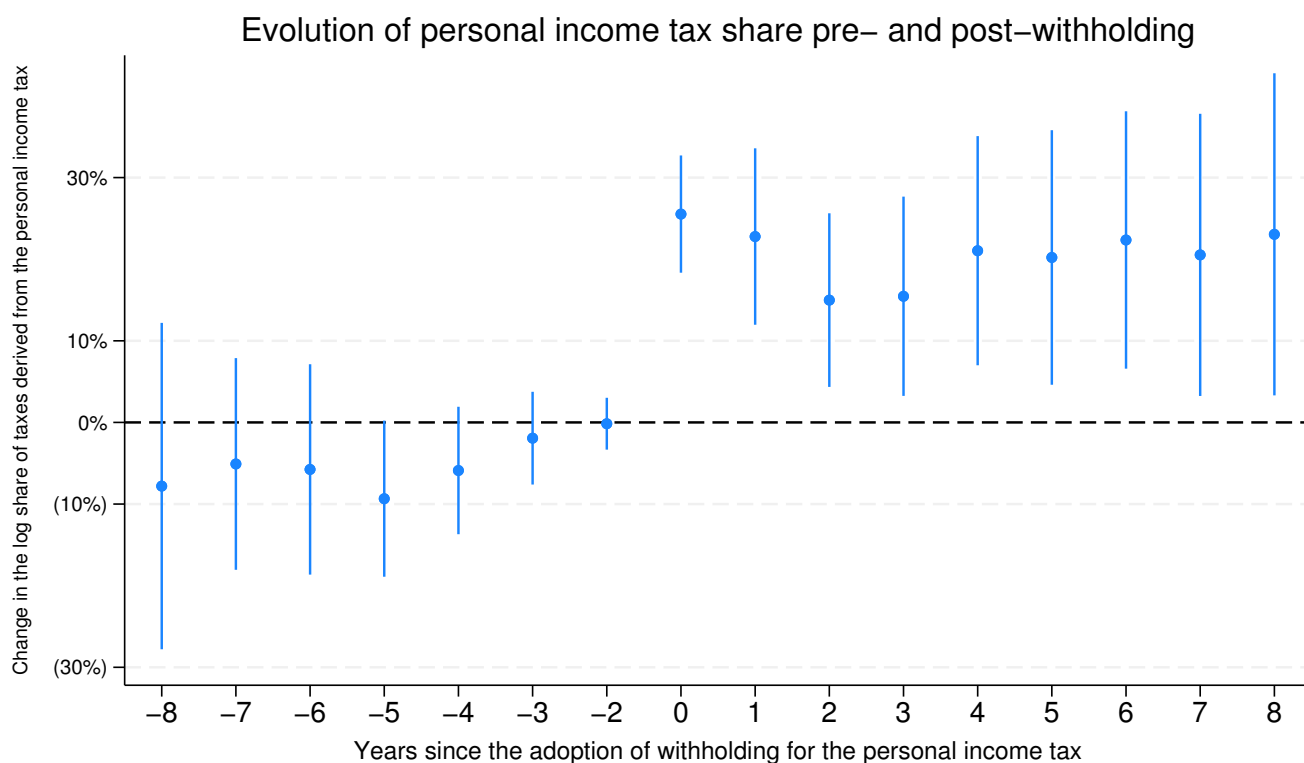
RC9: Splitting the sample based on the stringency of the balanced budget requirement

(BBR) As suggested by Figure 2, a possible reason for why taxes went up without a similar increase in expenses is the leveling off of state debt. The ability of states to borrow money in order to balance their budgets is influenced by whether the state has a BBR or not and how strict such a requirement is. To dig deeper into any possible heterogeneity of the effects of withholding, we split the sample into states classified by the Advisory Commission on Intergovernmental Relations (ACIR) as having a strict BBR or not.⁷⁵ We find that regardless of the presence or absence of a strict BBR, this fiscal institution does not affect our estimates of the effects of introducing withholding. For example, in the specification that includes all controls and state-specific time trends, the estimated effect of withholding is 0.321 (0.284) in states with (without) a strict BBR and those coefficients are not different from each other (p-value = 0.7434). Likewise, the estimated increase in the share of taxes collected from the personal income tax equals 0.299 (0.219) in states with (without) a strict BBR and those coefficients are not different from each other (p-value = 0.4845).

The results in cols. (1) and (2) of Table 6 provide robust evidence that the adoption of withholding resulted in a large and meaningful increase in personal income tax revenues and omitted variables, including pre-existing trends, are unlikely to be driving our results. In contrast, to the robustness of the findings involving the personal income tax, cols. (3) and (4) indicate more mixed results for all other taxes; although the coefficient on the withholding dummy is generally positive, it never attains statistical significance. Finally, given those sets of findings, it is unsurprising that we find robust increases in the share of taxes derived from the personal income tax when we examine that variable directly in cols. (5) and (6). Taken as a whole, the results from this table are most in line with Scenario 4 and indicate the presence of a positive demand shock as they confirm that personal income tax revenues went up after the introduction of withholding and other (non personal income) taxes were not reduced. Our view that there was a positive demand shock is also supported by the results in Table A.2 which replicates all of these robustness checks for the top personal income tax rate and the corporate income tax rate and confirms an increase of about 15 percent for both rates. Thus, increases in compliance following the introduction of withholding that led to higher collections did not lead to offsetting changes to other taxes or tax rates; instead, all evidence points to states increasing tax rates further in order to collect more revenues.

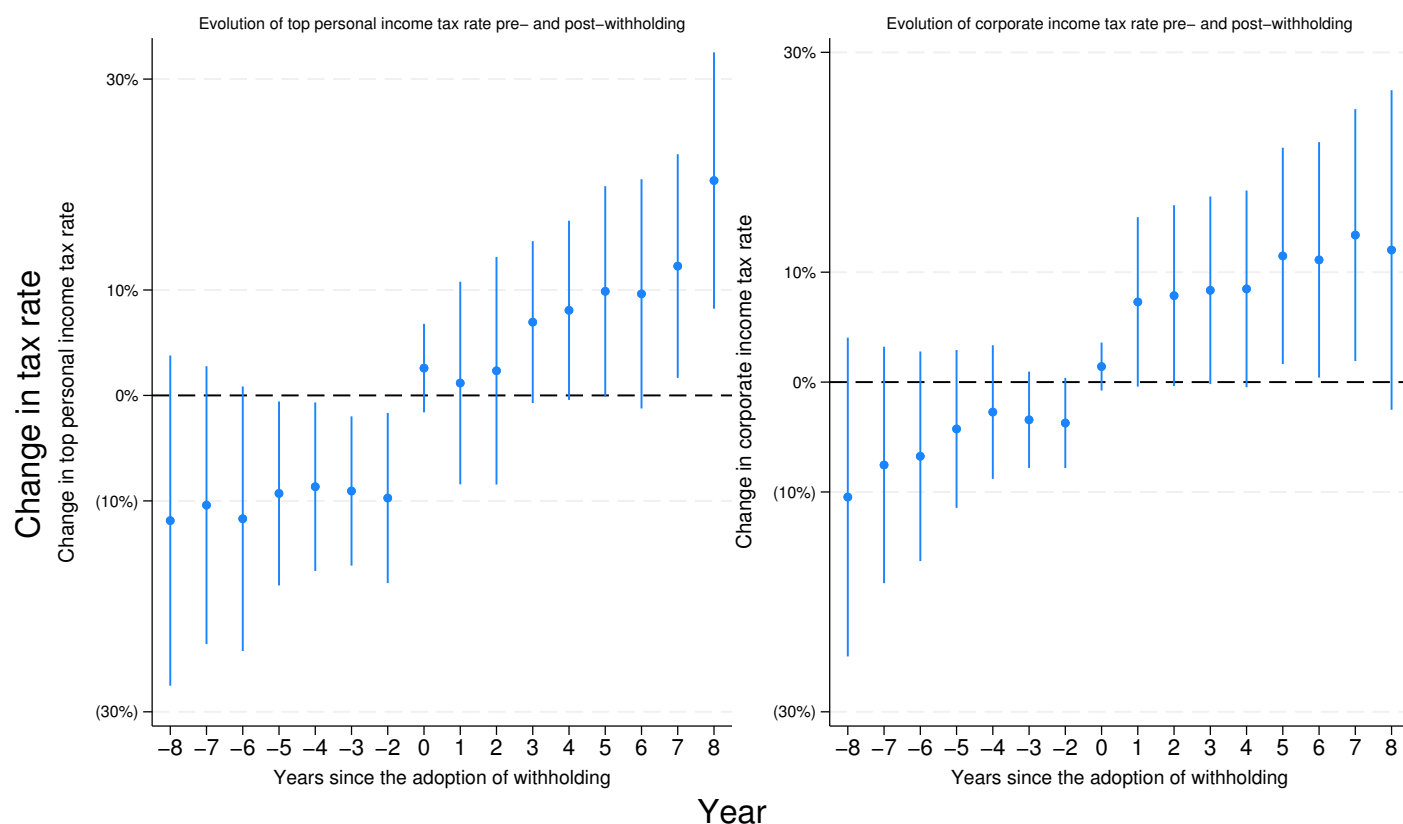
⁷⁵The data on BBRs are drawn from Bohn and Inman (1996), with the original source being a 1987 ACIR report. Of the 29 states in the sample, Bohn and Inman (1996) classify 16 states as having the highest ACIR stringency index of 10 and the remaining 13 states as having a stringency index lower than 10.”

Figure A.1: Estimated percentage change in share of taxes derived from the personal income tax for states adopting withholding after having introduced a state income tax, 95% CI



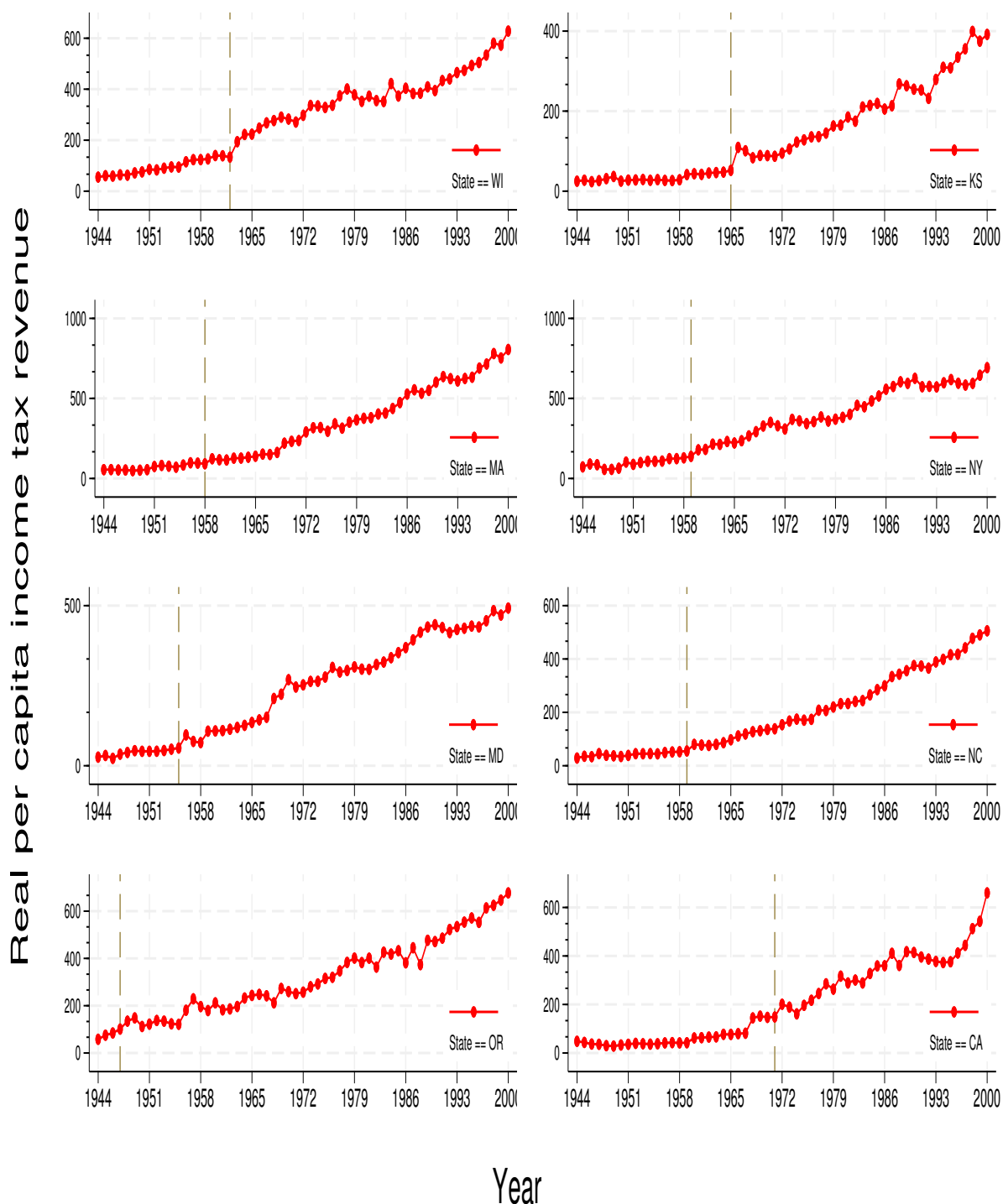
The figure plots the coefficients, along with 95 percent confidence intervals, from a regression of the log share of taxes derived from the personal income tax on a set of dummy variables for years prior to and after the adoption of withholding, state and year fixed effects, state-specific linear time trends, and all socio-economic controls, with standard errors clustered at the state level.

Figure A.2: Evolution of the top personal income tax rate and corporate income tax rate for states adopting withholding after having introduced a state income tax, 95% CI



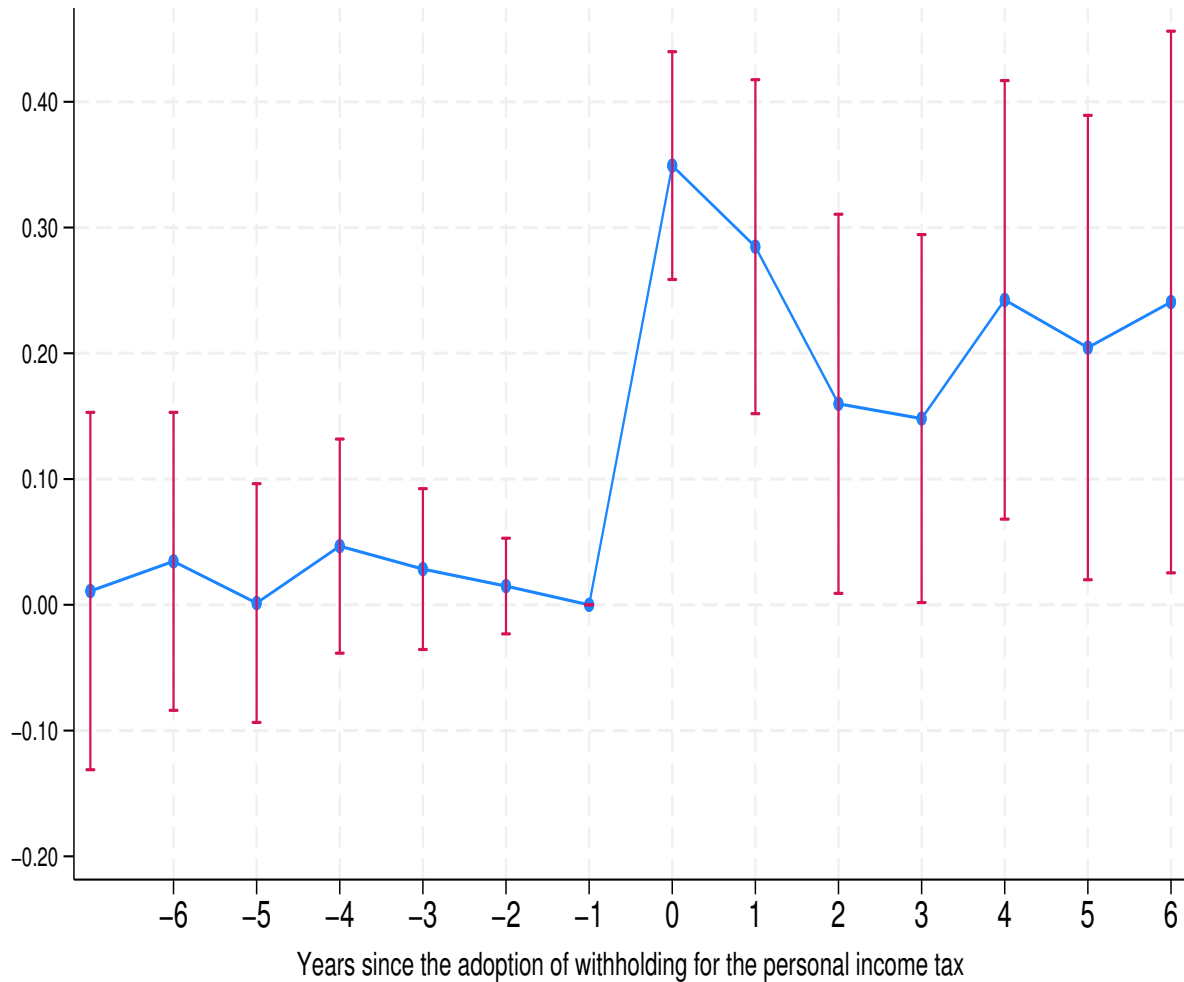
The left panel plots the coefficients, along with 95 percent confidence intervals, from a regression of the log top personal income tax rate on a set of dummy variables for years prior to and after the adoption of withholding, state and year fixed effects, state-specific linear time trends, and all socio-economic controls, with standard errors clustered at the state level. The right panel is generated in an identical manner but examines the evolution of the corporate income tax rate.

Figure A.3: Trends in personal income tax revenues for 8 states from across the U.S. for our entire sample period (1944–2000)



Note: The states chosen belong to the four different Census regions of the country: Wisconsin (WI) and Kansas (KS) are in the Midwest, Massachusetts (MA) and New York (NY) are in the Northeast, Maryland (MD) and North Carolina (NC) are in the South, and Oregon (OR) and California (CA) are in the West.

Figure A.4: Event-study plot using an estimator that is robust to heterogeneous treatment effects



Notes: The figure plots the coefficients, along with the 95 percent confidence intervals, from a regression of the log of real per capita income tax revenue on a set of dummy variables, state and year fixed effects, and state-specific linear time trends with standard errors clustered at the state level. This is drawn using the Stata module, “did_multiplegt” developed by de Chaisemartin and D’Haultfoeuille (2020) and uses 5,000 bootstraps.

Table A.1: Estimates of the effects of withholding on the personal income tax on balanced panels of varying lengths

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Log of real per capita personal income tax revenues					
Panel A: Estimated on a sample with at least 5 years of data pre- and post-adoption of withholding					
Withholding dummy	0.358*** (0.0790)	0.288*** (0.0731)	0.300*** (0.0804)	0.313*** (0.0834)	0.285*** (0.0672)
Observations	1596	1596	1596	1596	1596
Number of states included	28	28	28	28	28
Panel B: Estimated on a sample with at least 7 years of data pre- and post-adoption of withholding					
Withholding dummy	0.332*** (0.0807)	0.268*** (0.0755)	0.292*** (0.0843)	0.291*** (0.0853)	0.274*** (0.0694)
Observations	1539	1539	1539	1539	1539
Number of states included	27	27	27	27	27
Panel C: Estimated on a sample with at least 9 years of data pre- and post-adoption of withholding					
Withholding dummy	0.294*** (0.0753)	0.241*** (0.0751)	0.257*** (0.0815)	0.256*** (0.0834)	0.249*** (0.0698)
Observations	1482	1482	1482	1482	1482
Number of states included	26	26	26	26	26
Panel D: Estimated on a sample with at least 12 years of data pre- and post-adoption of withholding					
Withholding dummy	0.288*** (0.0787)	0.229*** (0.0764)	0.239*** (0.0848)	0.231** (0.0845)	0.241*** (0.0706)
Observations	1311	1311	1311	1311	1311
Number of states included	23	23	23	23	23
Panel E: Estimated on a sample with at least 13 years of data pre- and post-adoption of withholding					
Withholding dummy	0.244** (0.0941)	0.192* (0.0963)	0.174 (0.104)	0.162 (0.104)	0.208** (0.0911)
Observations	1083	1083	1083	1083	1083
Number of states included	19	19	19	19	19
Panel F: Estimated on a sample with at least 14 years of data pre- and post-adoption of withholding					
Withholding dummy	0.302** (0.117)	0.263** (0.114)	0.250** (0.116)	0.254** (0.102)	0.259** (0.102)
Observations	1026	1026	1026	1026	1026
Number of states included	18	18	18	18	18
Panel G: Estimated on a sample with at least 16 years of data pre- and post-adoption of withholding					
Withholding dummy	0.307** (0.126)	0.268** (0.123)	0.264* (0.126)	0.275** (0.109)	0.275** (0.107)
Observations	969	969	969	969	969
Number of states included	17	17	17	17	17
State and year fixed effects	Y	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	N	Y
Socio-economic controls	N	N	Y	Y	Y
Controls capturing demand for tax revenue	N	N	N	Y	Y

Std. errors clustered by state in parentheses * significant at 10%; ** significant at 5%; *** significant at 1%. For complete notes, refer to Table 3.

Table A.2: Robustness checks on estimates of the effect of withholding on the top personal and the corporate income tax rate

Dependent variable:	(1)	(2)	(3)	(4)
Baseline Specification	Log of top personal income tax rate	Log of top personal income tax rate	Log of corporate income tax rate	
	0.152** (0.0674)	0.134*** (0.0479)	0.136*** (0.0449)	0.110** (0.0410)
RC1: Pre-trend estimates (falsification exercise: only including pre-adoption years)	-0.0313 (0.0294)	-0.0157 (0.0253)	-0.0469 (0.0286)	-0.0438 (0.0275)
RC2: Ignoring transitory shocks in years 1 & 2 following adoption	0.197*** (0.0708)	0.166*** (0.0508)	0.161*** (0.0505)	0.132** (0.0491)
RC3: Including state-specific quadratic time trends	0.0851* (0.0447)	0.0927** (0.0398)	0.109** (0.0462)	0.109** (0.0459)
RC4: Introducing region-by-year FE	0.169** (0.0745)	0.141** (0.0530)	0.177*** (0.0602)	0.132*** (0.0403)
RC5: Introducing division-by-year FE	0.162 (0.101)	0.123* (0.0631)	0.196** (0.0881)	0.121** (0.0545)
RC6.1: Weighted regressions with contemporaneous state population as weights	0.145** (0.0646)	0.131*** (0.0468)	0.129*** (0.0455)	0.107** (0.0412)
RC6.2: Weighted regressions with state population in 1944 as weights	0.147** (0.0642)	0.131*** (0.0458)	0.130*** (0.0454)	0.107** (0.0412)
RC7: Use of a random effects specification	0.141* (0.0763)	0.147*** (0.0521)	0.179** (0.0732)	0.153*** (0.0579)
RC8: Controlling for the presence of a federal-state information sharing agreement	0.153** (0.0673)	0.134*** (0.0477)	0.136*** (0.0437)	0.112*** (0.0394)
RC9.1: For states with strict balanced budget requirement (ACIR stringency index = 10)	0.0485 (0.0714)	0.0838 (0.0691)	0.132 (0.0771)	0.131* (0.0734)
RC9.2: For states with no strict balanced budget requirement (ACIR stringency index < 10)	0.254** (0.0924)	0.157** (0.0564)	0.130** (0.0594)	0.0429 (0.0373)
State and year fixed effects	Y	Y	Y	Y
State-specific linear time trends	Y	Y	Y	Y
Socio-economic controls	N	Y	N	Y
Controls capturing demand for tax revenue	N	Y	N	Y

Standard errors clustered by state in parentheses * p < 0.10, ** p < 0.05, *** p < 0.01.

Results in this table are variations on the specifications reported in columns (2) and (5) of Table 7. The top row of the current table is the baseline specification. Except where noted, all specifications are estimated using an annual, state-level panel of data for the years 1944–2000. A list of the states included in each Census region and division is available at https://www2.census.gov/geo/maps-data/maps/reg_div.txt. Region-by-year FE in RC4 and division-by-year FE in RC5 are for the four Census regions and nine Census divisions respectively. State and year fixed effects and state-specific linear time trends are included in all specifications. Columns (2) and (4) include socio-economic controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas as well as demand-side factors, viz. female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding.

Table A.3: Estimates of the effects of withholding on state expenditures

	(1)	(2)	(3)	(4)	(5)
Dependent variable: Panel A: Real general expenditures as a percentage of real personal income					
Withholding dummy	-0.0913 (0.251)	0.0758 (0.202)	-0.0285 (0.224)	0.0991 (0.244)	0.131 (0.238)
Dependent variable: Panel B: Real spending on education as a percentage of real personal income					
Withholding dummy	0.0180 (0.165)	-0.0138 (0.137)	0.0727 (0.165)	0.0263 (0.151)	0.0425 (0.154)
Dependent variable: Panel C: Real spending on roads as a percentage of real personal income					
Withholding dummy	0.0636 (0.115)	0.0111 (0.108)	0.0594 (0.107)	0.0657 (0.0945)	0.0591 (0.0773)
Dependent variable: Panel D: Real spending on welfare as a percentage of real personal income					
Withholding dummy	-0.0220 (0.141)	-0.0210 (0.127)	-0.0277 (0.130)	0.0525 (0.147)	-0.0402 (0.0990)
Dependent variable: Panel E: Real spending on health as a percentage of real personal income					
Withholding dummy	0.0177 (0.0370)	-0.0246 (0.0183)	-0.0204 (0.0330)	-0.0173 (0.0321)	-0.0408* (0.0232)
Dependent variable: Panel F: Real spending on natural resources as a percentage of real personal income					
Withholding dummy	-0.00499 (0.0323)	-0.0102 (0.0280)	-0.00794 (0.0293)	-0.00907 (0.0283)	-0.00924 (0.0272)
Dependent variable: Panel G: Real spending on public safety as a percentage of real personal income					
Withholding dummy	-0.0301 (0.0204)	-0.0207** (0.00797)	-0.0390*** (0.0141)	-0.0315** (0.0127)	-0.0237*** (0.00688)
Dependent variable: Panel H: Real spending on other categories as a percentage of real personal income					
Withholding dummy	-0.133 (0.118)	0.155 (0.144)	-0.0655 (0.115)	0.0125 (0.109)	0.143 (0.122)
Observations (for all panels)	1653	1653	1653	1653	1653
State and year fixed effects	Y	Y	Y	Y	Y
State-specific linear time trends	N	Y	N	N	Y
Socio-economic controls	N	N	Y	Y	Y
Controls capturing demand for tax revenue	N	N	N	Y	Y

Standard errors clustered by state in parentheses *p < 0.10, ** p < 0.05, *** p < 0.01.

State and year fixed effects are included in all specifications; state-specific linear time trends are included in columns (2) and (5). Columns (3), (4), and (5) include controls for state population, real per capita income, share of farm income, and the share of interest and dividend income in state personal income, the percentage of state population that is black, young (less than 15), elderly (65+), and lives in urban areas. Columns (4) and (5) include female labor force participation, the share of income going to the top 1 percent of the population, and the average vote share of Democrats in all state-wide races. Fiscal and socio-economic variables are in logs. Fiscal variables are in per capita terms. The regressions are estimated using the sample of 29 states that had a broad-based personal income tax as of 1944 and then subsequently introduced withholding.